

Human immunodeficiency virus-1 induces host genomic R-loop and preferentially integrates its genome near the R-loop regions

Reviewed Preprint

v2 • November 11, 2024

Revised by authors

Reviewed Preprint

v1 • June 21, 2024

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eLife Assessment

This study presents two main findings regarding HIV-1 genomic integration. The first, based on **convincing** evidence in primary cell models, is that HIV-1 induces R loop formation, though the viral driver of this process remains undefined. The second, based on model cell systems with limited physiological relevance to HIV-1, is that a portion of HIV-1 genomes integrates in the vicinity of where R loops form. This finding has the potential to offer **fundamental** insight into HIV-1 integration, but the strength of the presented evidence was viewed as **incomplete** and needing additional validation by more direct experimental methods in order to understand what the mechanistic relationship between the formation of R loops and HIV-1 integration is.

<https://doi.org/10.7554/eLife.97348.2.sa3>

Abstract

Although HIV-1 integration sites favor active transcription units in the human genome, high-resolution analysis of individual HIV-1 integration sites has shown that the virus can integrate into a variety of host genomic locations, including non-genic regions. The invisible infection by HIV-1 integrating into non-genic regions, challenging the traditional understanding of HIV-1 integration site selection, is more problematic because they are selected for preservation in the host genome during prolonged antiretroviral therapies. Here, we showed that HIV-1 integrates its viral genome into the vicinity of R-loops, a genomic structure composed of DNA– RNA hybrids. VSV-G-pseudotyped HIV-1 infection initiates the formation of R-loops in both genic and non-genic regions of the host genome and preferentially integrates into R-loop-rich regions. Using a HeLa cell model that can independently control transcriptional activity and R-loop formation, we demonstrated that the exogenous formation of R-loops directs HIV-1 integration-targeting sites. We also found

that HIV-1 integrase proteins physically bind to the host genomic R-loops. These findings provide novel insights into the mechanisms underlying retroviral integration and the new strategies for antiretroviral therapy against HIV-1 latent infection.

Introduction

Retroviruses cause permanent host infections by integrating their reverse-transcribed viral genomes into the host genome. Retroviral integration considerably affects a wide range of biological phenomena, including the persistence of fatal human diseases and the shaping of metazoan evolution (1). Human immunodeficiency virus (HIV)-1 is a representative retrovirus underlying the global burden of acquired immune deficiency syndrome (AIDS) (2). The chromosomal landscape of HIV-1 integration has been shown to influence proviral gene expression, the persistence of integrated proviruses, and prognosis in antiretroviral therapy (3–5). Integration into the host genome is not random and displays distinct preferences for gene-dense regions, where active transcription occurs (6), by interacting with host factors, such as transcription activators, epigenetic marker binding proteins, and super enhancers (7–13). However, transcriptional activity is not the sole determinant of the HIV-1 integration site landscape (10). For instance, some highly expressed genes do not necessarily correspond to a high level of integration in PBMCs infected in culture with HIV (14). Despite their lower probability of integration, HIV-1 proviruses are observed in non-genic regions of the genomes of infected individuals (4). This indicates the possibility of there being an undiscovered mechanism or determinant that constitutes the correct genomic environment for HIV-1 integration.

An R-loop is a three-stranded nucleic acid structure that comprises a DNA–RNA hybrid and a displaced strand of DNA, and has long been considered a transcription byproduct (15, 16). R-loops in cellular genomes are enriched in actively transcribed genes as they occur naturally during transcription (15, 17). However, R-loop formation is not limited to gene body regions and is widespread in the genome (15). As a result of *in trans* R-loop formation, R-loops are also abundant in non-genic regions, such as intergenic regions and repetitive sequences, including transposable elements, centromeres, and telomeres (15, 18–20), independent of the transcriptional activity of the genes harboring the R-loops. Although R-loops have been identified as critical intermediates and regulators of a number of biological processes (15, 16, 21), the dynamics and the role played by cellular R-loops in pathological contexts remain unclear.

R-loops are important contributors to molding the genomic environment and spatial organization of the cellular genome, and can potentially play a novel role in host-pathogen interaction. In the cellular genome, R-loops relieve superhelical stresses and are often associated with open chromatin marks and active enhancers (22, 23), which are also distributed over HIV-1 integration sites (6, 9, 10). In transcription-induced R-loop formation, a guanine-quadruplex (G4) structure can be generated in the non-template DNA strand of the R-loop (24). A recent study showed that G4 DNA can influence both productive and latent HIV-1 integration (25). In addition, R-loops are prevalent non-canonical B-form DNA structures (26) and intermediates between the B-form DNA and A-form RNA conformations (27). It has recently been reported that the B-to-A transition in the target DNA occur during retroviral integration (27, 28). Accumulating evidence suggests that host genomic R-loops are undiscovered host factors in the HIV-1 integration site selection mechanism that dynamically interact with the host genomic environment.

Here, we showed a notable role for R-loops in the interaction between HIV-1 and its host, specifically in HIV-1 integration. VSV-G-pseudotyped HIV-1-infection induces host cellular R-loop formation, and the R-loop rich regions of the host genome are preferred for HIV-1 integration in

diverse cell types. HIV-1 integrase proteins showed considerable binding affinity for nucleic acid substrates containing R-loop structures. Our results suggest that R-loops are important components of the host genomic environment for HIV-1 integration site determination.

Results

Host genomic R-loops accumulate by VSV-G-pseudotyped HIV-infection in diverse cell types

To investigate the relationship between HIV-1 infection and host cellular R-loops, we first analyzed R-loop dynamics in different types of cells infected with HIV-1 at early post-infection time points using DNA–RNA immunoprecipitation followed by cDNA conversion coupled with high-throughput sequencing (DRIPc-seq) using a DNA–RNA hybrid-specific binding antibody, anti-S9.6 (29). HeLa cells, primary CD4⁺ T cells isolated from two individual donors, and the CD4⁺/CD8[−] T cell lymphoma Jurkat cell line were infected with the VSV-G-pseudotyped HIV-1-EGFP in a time-course manner. The infected cells were harvested at the same post-seeding time point but at 0, 3, 6, and 12 h post infection (hpi) for DRIPc-seq library construction (Fig. 1A and S1A–C Fig.). Our DRIPc-seq analysis yielded loci-specific R-loop signals at the referenced R-loop-positive loci (RPL13A and CALM3) and an R-loop-negative locus (SNRPN) (29), which were both strand-specific and highly sensitive to pre-immunoprecipitation *in vitro* RNase H treatment in HeLa cells, CD4⁺, and Jurkat T cells (Table S1–3). Notably, the number of DRIPc-seq peaks mapped to the human reference genome increased markedly during the early post infection with HIV-1 (3 and 6 hpi for HeLa cells and 6 and 12 hpi for CD4⁺ and Jurkat T cells; Fig. 1B). Most of the peaks that mapped in cells harvested at 0 hpi were commonly found in all other samples, but a significant numbers of unique peaks were observed after infection (Fig. 1C).

In addition to the DRIPc-seq data analysis, we used different biochemical approaches to examine R-loop accumulation after HIV-1 infection in HeLa cells. R-loop accumulation in HIV-1-infected cells was observed using DNA–RNA hybrid dot blots with the anti-S9.6 antibodies (Fig. 1D). Dot intensity significantly increased upon HIV-1 infection at 6 hpi and the enhanced R-loop signals on the dot blots of HIV-1-infected cells were highly sensitive to *in vitro* treatment with RNase H (Fig. 1D). This result is highly consistent with our DRIPc-seq data analysis of HIV-1-infected HeLa cells. Subsequently, we observed HIV-1-induced R-loops using an immunofluorescence assay by probing HIV-1-infected or non-infected control cells with S9.6 antibody at 6 hpi (Fig. 1E). After subtracting the nucleolar signal, the nuclear fluorescence signal associated with R-loops was significantly enhanced in cells infected with HIV-1 (Fig. 1E). We validated and quantified HIV-1-infection induced R-loop formation on the host genome in a genome-site-specific manner by using DRIP followed by real-time polymerase chain reaction (DRIP-qPCR). In this experiment, the S9.6 signal was determined for three and two HIV-1-induced-R-loop-positive (P1, P2, and P3) and –negative regions (N1 and N2), respectively, which were defined by DRIPc-seq data analysis (S2A–E Fig.). We detected significantly increased R-loop signals that were highly sensitive to RNase H treatment of pre-immunoprecipitates in the P1, P2, and P3 regions of HIV-1-infected cells at 6 hpi compared to those in the cells harvested at 0 hpi (S3A Fig.). However, the HIV-1-induced R-loop-negative regions N1 and N2, did not show significant R-loop accumulations (S3A Fig.).

Importantly, the R-loop signal was enriched even in cells infected with HIV-1 when the reverse transcription or integration of HIV-1 was blocked by enzyme inhibitors, such as nevirapine (NVP) or raltegravir (RAL) (S3B and S3C Fig.). This result indicates that the enrichment of R-loop signals in cells originates from the host genome, but not from DNA–RNA hybrid formation during the viral life cycle or transcriptional burst from integrated HIV-1 proviruses. In addition, we confirmed that nearly 100% of DRIPc-seq reads from HIV-1-infected HeLa, CD4⁺, and Jurkat T cells were aligned to the host cellular genome, but not to that of HIV-1 (S3D Fig.). Together, these data demonstrated

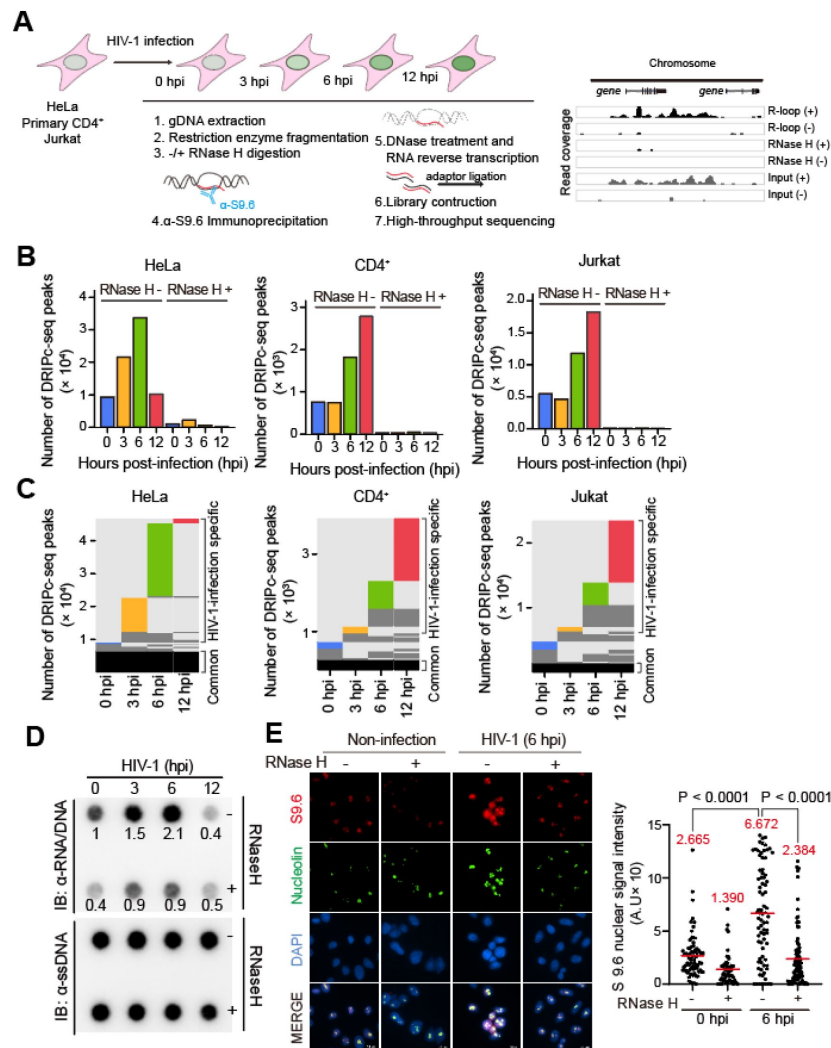


Fig 1.

HIV-1 infection induces genomic R-loop accumulation in cells at early post-infection.

(A) Summary of experimental design for DRIPc-seq in HeLa cells, primary CD4⁺ T cells and Jurkat cells infected with HIV-1. (B) Bar graphs indicating DRIPc-seq peak counts for HIV-1-infected HeLa cells, primary CD4⁺ T cells and Jurkat cells harvested at the indicated hours post infection (hpi). Pre-immunoprecipitated samples were untreated (-) or treated (+) with RNase H, as indicated. Each bar corresponds to pooled datasets from two biologically independent experiments. (C) All genomic loci overlapping a DRIPc-seq peak from HIV-1 infected HeLa cells, primary CD4⁺ T cells and Jurkat cells in at least one sample are stacked vertically; the position of each peak in a stack is constant horizontally across samples. Each hpi occupies a vertical bar, as indicated. Each bar corresponds to pooled datasets from two biologically independent experiments. Common peaks for all samples are represented in black, and in dark gray for those common for at least two samples. The lack of a DRIP signal over a given peak in any sample is shown in light gray. The sample-unique peaks are colored blue, yellow, green, and red at 0, 3, 6, and 12 hpi, respectively. (D) Dot blot analysis of the R-loop in gDNA extracts from HIV-1 infected HeLa cells with MOI of 0.6 harvested at the indicated hpi. gDNAs were probed with anti-S9.6. gDNA extracts were incubated with or without RNase H in vitro before membrane loading (anti-RNA/DNA signal). Fold-induction was normalized to the value of harvested cells at 0 hpi by quantifying the dot intensity of the blots and calculating the ratios of the S9.6 signal to the total amount of gDNA (anti-ssDNA signal). (E) Representative images of the immunofluorescence assay of S9.6 nuclear signals in HIV-1 infected HeLa cells with MOI of 0.6 harvested at 6 hpi. The cells were pre-extracted of cytoplasm and co-stained with anti-S9.6 (red), anti-nucleolin antibodies (green), and DAPI (blue). The cells were incubated with or without RNase H in vitro before staining with anti-S9.6 antibodies, as indicated. Quantification of S9.6 signal intensity per nucleus after nucleolar signal subtraction for the immunofluorescence assay. The mean value for each data point is indicated by the red line. Statistical significance was assessed using one-way ANOVA ($n > 53$).

that HIV-1 infection induce host genomic R-loop enrichment at early post-infection. This also suggests that HIV-1 infection regulates R-loop dynamics in host cells, possibly by inducing either new R-loop formation or stabilizing existing R-loops in the host cellular genome.

R-loops accumulation after HIV-1 infection are widely distributed in both genic and non-genic regions

To investigate the distribution of cellular genomic R-loops during HIV-1 infection, we conducted a genome-wide analysis of the DRIPc-seq data. Unique DRIPc-seq peaks observed after HIV-1 infection were numerous and relatively long (Fig. 2A). This suggests that the R-loops induced by HIV-1 infection occupy a genomic region larger than that of the R-loops without HIV-1 infection. We observed a significant accumulation of R-loops over diverse genomic compartments at the hpi of HIV-1-infection induced R-loop formation (Fig. 2B). The presence of R-loops is often correlated with high transcriptional activity, and we found a significantly high proportion of DRIPc-seq peaks enrichment upon HIV-1 infection in the gene body regions (Fig. 2B). However, we also observed enrichment of HIV-1-infection induced DRIPc-seq peaks proportions mapped to intergenic or repeat regions, including short interspersed nuclear elements (SINEs), long interspersed nuclear elements (LINEs), and long terminal repeat (LTR) retrotransposons, where transcription is typically repressed (Fig. 2B). In addition, the proportion of DRIPc-seq peaks mapped to various genomic compartments remained consistent over the hours following HIV-1 infection (Fig. 2C). This suggests that HIV-1 infection does not induce R-loop enrichment at specific genomic features, but that R-loops accumulation after HIV-1 infection is widely distributed. Although the expression of repetitive elements is mostly repressed during normal cellular activities, HIV-1 infection could activate endogenous retroviral promoters (30, 31). To investigate whether R-loop induction in gene-silent regions is associated with transcriptome changes during HIV-1 infection, we performed RNA sequencing (RNA-seq) of HIV-1-infected HeLa cells at 0, 3, 6, and 12 hpi. Consistent with previous reports, we observed an increase in the expression levels of repetitive elements at later time points post-infection (S4A Fig. 12 hpi). In contrast, we found that the expression levels of SINEs, LINEs, and LTRs were even lower at both 3 and 6 hpi compared with those at 0 hpi whereas HIV-1-induced R-loops were significantly accumulated compared with those at 0 hpi (S4A Fig.). We further examined the expression profiles of genes containing the R-loop in HeLa cells. The expression profile of genes harboring HIV-1-induced R-loops in their gene bodies showed very weak correlations with the signals of DRIPc-seq peaks at 3 hpi (Pearson's $r = 0.21$, $P = 1.08 \times 10^{-84}$, Fig. 2D) and at 6 hpi (Pearson's $r = -0.34$, $P = 2.40 \times 10^{-228}$, Fig. 2D), which implies that the unique R-loop peaks upon HIV-1 infection do not engage in a transcriptional burst. In agreement with our DRIPc-seq and global RNA-seq data analysis, the expression level of the genes harboring HIV-1-infection induced R-loops, which were quantified by DRIP-qPCR (S3A Fig.), were not significantly affected by HIV-1 infection (S4B Fig. and Table S4). Together, our data demonstrate that host cellular R-loop accumulation upon HIV-1 infection is widely distributed in both genic and non-genic regions and is not necessarily correlate with the expression levels of the genes harboring the R-loops.

HIV-1 integration sites are enriched at systemically induced sequence-specific R-loop regions in cell model

HIV-1 infection is completed by integrating the viral genome into the host's through dynamic interactions with the host genome (32). In addition, as HIV-1 infection induces R-loop accumulation at early post infection hours, when the HIV-1 genome are imported into the nucleus and integration may initiate (33–35), we hypothesized that host genomic R-loops play a role in HIV-1 integration and possibly in integration site selection. To systemically and directly assess the relationship between host genomic R-loops and HIV-1 integration in a genome site-specific manner, we adapted and modified an elegantly designed episomal system that induces sequence-specific R-loops through DOX-inducible promoters (17). To most closely mimic the presence of

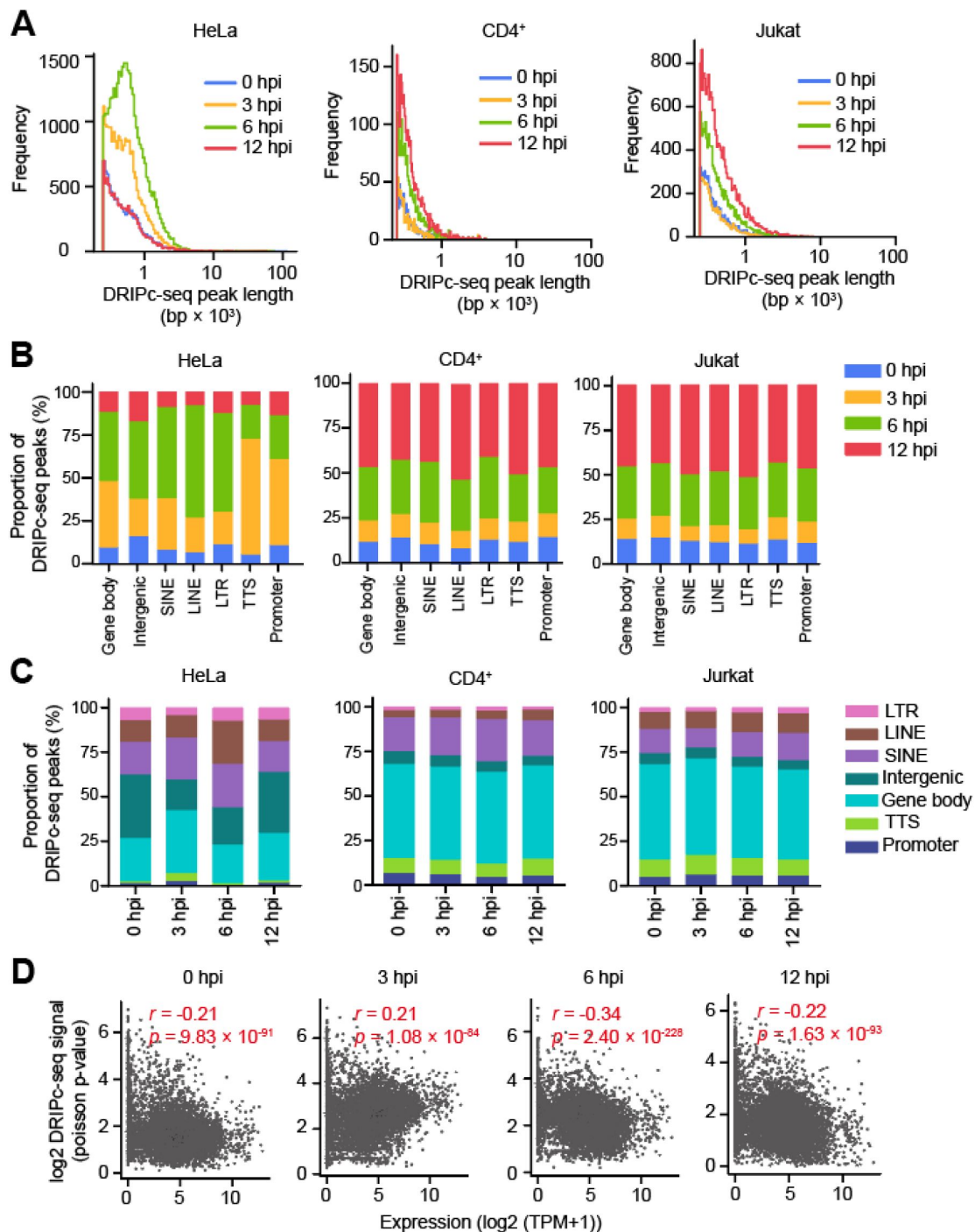


Fig. 2.

HIV-1-induced R-loops are enriched at both transcriptionally active and silent regions.

(A) Distribution of DRIPc-seq peak lengths for HIV-1-infected HeLa cells, primary CD4⁺ T cells and Jurkat cells harvested at the indicated time points (blue, 0 hpi; yellow, 3 hpi; green, 6 hpi; red, 12 hpi). (B) Stacked bar graphs indicating the proportion of DRIPc-seq peaks mapped for HIV-1-infected HeLa cells, primary CD4⁺ T cells and Jurkat cells harvested at the indicated hpi over different genomic features. (C) Stacked bar graphs indicating the proportion of DRIPc-seq peaks mapped to indicated genomic compartments for HIV-1-infected HeLa cells, primary CD4⁺ T cells and Jurkat cells harvested at the 0, 3, 6, and 12 hpi. (D) Correlation between gene expression and DRIPc-seq signals of HIV-1-infected HeLa cells with MOI of 0.6 harvested at the indicated hpi. Statistical significance was assessed using Pearson's r and p -values.

the R-loop in the host cellular genome, we subcloned the R-loop-forming portion of the mouse gene encoding AIRN (mAIRN) (18 [↗](#)) or the non-R-loop-forming ECFP sequence (17 [↗](#)) with a DOX-inducible promoter into the piggyBac transposon vector and co-expressed piggyBac transposase in HeLa cells. The mAIRN gene sequence possesses a high GC skew that causes R-loop structure formation upon transcription by hybridizing newly synthesized RNA back to the template DNA strand, and the non-template DNA strand to remains looped out in a single-stranded form (18 [↗](#), 36 [↗](#)). In contrast, the ECFP sequence possesses a low GC skew that does not form a stable R-loop at the site of transcription. Thus, it was used as an R-loop-forming negative sequence. These R-loop forming (mAIRN) or non-R-loop forming (ECFP) sequences are nonhuman sequences. Therefore, our cell model allowed us to induce and quantify R-loop formation at designated genomic regions and distinguish R-loop formation from endogenous R-loops on the cellular genome, which are not sequence-specific and impossible to control for induction. Moreover, using this system we can quantify R-loop-associated site-specific HIV-1 integration events at designated regions, which can also be distinguished from HIV-1 integration events at endogenous host genomic loci. We designated the pool of cells with the R-loop forming sequence (mAIRN) inserted into its genome as “pgR-rich (piggyBac R-loop rich)” cell line and the pool of cells with the non-R-loop forming sequence (ECFP) inserted into its genome as “pgR-poor (piggyBac R-loop poor)” cell line (**Fig. 3A** [↗](#)).

A similar number of the copies of the piggyBac transposon were successfully delivered to the genome of each cell line (S5A Fig.) and DOX treatment strongly induced the transcriptional activity of mAIRN or ECFP without affecting the transcription of endogenous loci in either cell line (S5B and S5C Fig.). Although the transcription of mAIRN or ECFP was strongly induced by DOX treatment, the activity did not exceed that of the endogenous loci in either cell line (S5D and S5E Fig.). Although the two cell lines showed comparable levels of DOX-inducible transcriptional activity at the designated sequences (**Fig. 3B** [↗](#)), only the pgR-rich cells exhibited robust RNase H-sensitive stable R-loop formation upon DOX treatment (**Fig. 3C** [↗](#), mAIRN). In contrast, R-loops were weakly formed in the pgR-poor cells where a non-R-loop-forming sequence (ECFP) was inserted into the genome (**Fig. 3C** [↗](#), ECFP).

To examine whether the formation of ‘extra’ R-loops in the host genome influence HIV-1-infection in host cells, we infected both cell lines with VSV-G-pseudotyped HIV-1-luciferase viruses and examined the luciferase activity. Notably, we found that pgR-rich cells showed significantly higher luciferase activity only when R-loops were induced by DOX treatment, whereas pgR-poor cells showed comparable luciferase activity regardless of transcriptional activation by DOX treatment (**Fig. 3D** [↗](#)). We sequenced HIV-1 integration sites in HIV-1-infected pgR-poor and pgR-rich cells to directly quantify site-specific integration events in sequence-specific R-loop regions. Remarkably, integration events were significantly higher in pgR-rich cells only when R-loops were induced by DOX treatment (**Fig. 3E** [↗](#)). However, the number of HIV-1 integration-seq mapped reads within the non-R-loop forming sequence in pgR-poor cells remained very low, even after transcriptional activation by DOX treatment (**Fig. 3E** [↗](#)). HIV-1 integration was enriched in the vicinity of R-loop forming regions in the pgR-rich cell line upon DOX treatment, but the enrichment was not observed in pgR-poor cells that did not form stable R-loops, even after transcriptional activation by DOX treatment (**Figs. 3F** [↗](#) and **3G** [↗](#)). This cell-based R-loop-inducing system with independent control over transcription and R-loop formation enabled the direct measurement of HIV-1 integration events at defined R-loop regions. The results indicated that host genomic R-loops are preferred by HIV-1 integration. Moreover, our data suggest that transcriptional activity itself is not sufficient for HIV-1 integration site determination, but that the formation of R-loops accounts for HIV-1 integration site selection.

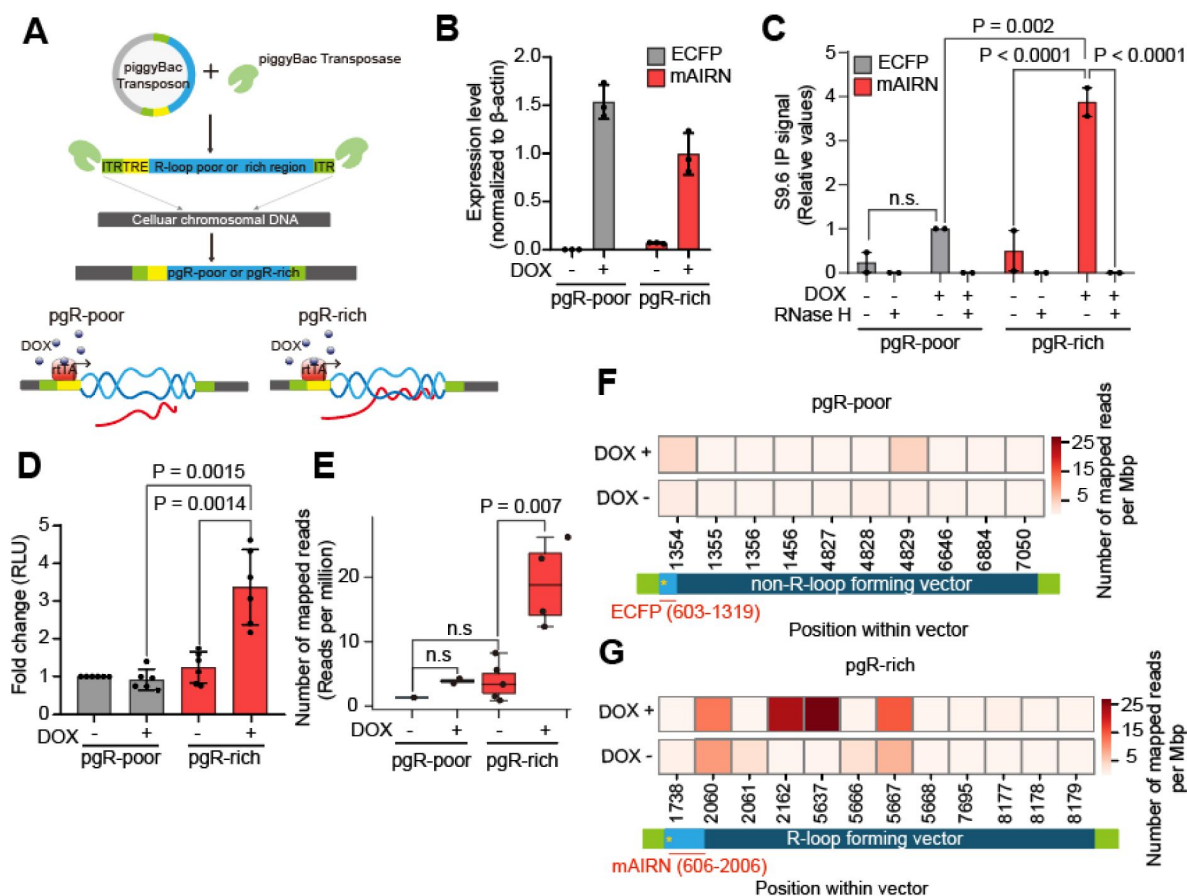


Fig. 3.

R-loop inducible cell line model directly addresses R-loop-mediated HIV-1 integration site selection.

(A) Summary of the experimental design for R-loop inducible cell lines, pgR-poor and pgR-rich. (B) Gene expression of ECFP (gray) and mAIRN (red), as measured using RT-qPCR in pgR-poor or pgR-rich cells. Where indicated, the cells were incubated with 1 μ g/ml DOX for 24 h. Gene expression was normalized relative to β -actin. Data are presented as the mean $\pm$ SEM, $n = 3$. (C) DRIP-qPCR using the anti-S9.6 antibody against ECFP and mAIRN in pgR-poor or pgR-rich cells. Where indicated, the cells were incubated with 1 μ g/ml DOX for 24 h. Pre-immunoprecipitated samples were untreated or treated with RNase H as indicated. Values are relative to those of DOX-treated (+) RNase H-untreated (-) pgR-poor cells. Data are presented as the mean $\pm$ SEM; statistical significance was assessed using two-way ANOVA ($n = 2$). (D) Bar graphs indicate luciferase activity at 48 hpi in pgR-poor or prR-rich cells infected with 100ng/p24 capsid antigen of luciferase reporter HIV-1 virus per 1×10^5 cells/mL. Data are presented as the mean $\pm$ SEM; P values were calculated using one-way ANOVA ($n = 6$). (E) Box graph indicating the quantified HIV-1 integration site sequencing read count across pgR-poor and pgR-rich transposon sequences in untreated (-) or DOX-treated (+) pgR-poor or pgR-rich cell line infected with 100ng/p24 capsid antigen of luciferase reporter HIV-1 virus per 1×10^5 cells/mL. Each bar corresponds to pooled datasets from three biologically independent experiments ($n = 3$). In each boxplot, the centerline denotes the median, the upper and lower box limits denote the upper and lower quartiles, and the whiskers denote the $1.5 \times$ interquartile range. Statistical significance was assessed using a two-sided Mann-Whitney U test. (F and G) Heat maps representing the number of HIV-1 integration-seq mapped read across pgR-poor (F) or pgR-rich (G) transposon sequence in untreated (-) or DOX-treated (+) pgR-poor (F) or pgR-rich (G) cell line. Each rectangular box corresponds to the pooled the number of HIV-1 integration-seq mapped read from three biologically independent experiments ($n = 3$) at the indicated position within pgR-poor (F) or pgR-rich (G) transposon vector. Each light blue box represents actual position of R-loop forming or non-R-loop forming sequence (ECFP or mAIRN) and the yellows stars indicate TRE promoter position within vector.

Host genomic R-loop regions are frequently targeted by HIV-1 integration

We attempted to further validate the relationship between R-loops and HIV-1 integration site selection by global analysis of HIV-1 integration sites in the endogenous genomic regions of HIV-1 infected host cells. We performed HIV-1 integration site sequencing in HIV-1 infected HeLa cells, CD4⁺, and Jurkat T cells, and analyzed the sequencing data combined with our DRIPc-seq data. We counted and compared the number of successfully integrated proviruses in the R-loop regions (the combined genomic regions within 30-kb windows centered on DRIPc-seq peaks from 0, 3, 6, and 12 hpi) to those in non-R-loop-forming regions (the total genomic regions outside of the 30-kb windows centered on DRIPc-seq peaks). Notably, we detected approximately three to four times more integration in the R-loop regions than in other genomic regions without R-loops in HeLa cells, CD4⁺, and Jurkat T cells (**Fig. 4A** [↗](#)). Notably, HIV-1 integration sites preferred the central and nearby areas of the R-loops regions (**Fig. 4B** [↗](#)). We observed biases in HIV-1 integration in HIV-1-induced R-loop-positive regions, P1-P3, where showed highly induced R-loop signals upon HIV-1 infection in DRIPc-seq analysis and DRIP-qPCR (**Fig. 4C** [↗](#)). In contrast, HIV-1 integration sites were not detected in the R-loop-negative regions N1 and N2 (**Fig. 4D** [↗](#)). Overall, our results from bioinformatics analysis using different types of naïve host cells infected with HIV-1 are consistent with the idea that the virus has a preference for targeting R-loop-forming regions for integration (**Fig. 3** [↗](#)), suggesting that R-loops are an important component of the host genomic environment for HIV-1 integration site determination.

HIV-1 integrase proteins physically interacts with R-loops on the host genome

The HIV-1 intasome is tethered to the host genome for viral cDNA integration. Intasomes consist of HIV-1 viral cDNA and the HIV-1 integrase proteins. We observed that HIV-1 was preferentially integrated into the R-loops regions in the host genome; thus, we hypothesized that the HIV-1 integrase protein could directly bind and be recruited to the genomic R-loops. To test this hypothesis, we investigated whether HIV-1 integrase proteins have a physical binding affinity for nucleic acid substrates with an R-loop structure. Although HIV-1 integrases are DNA and RNA binding proteins ([37](#) [↗](#), [38](#) [↗](#)), their ability to bind to a three-stranded nucleic acid structure that is composed of a DNA-RNA hybrid like R-loop has not been investigated. We carried out an *in vitro* protein-nucleic acid binding assay by electrophoretic mobility shift assay (EMSA) with Sso7d-tagged HIV-1 integrase (E152Q) recombinant proteins and diverse structures of nucleic acid substrates including R-loops and simple dsDNA duplexes. In this experiment, we used HIV-1 integrase protein with an active site amino acid substitution, E152Q, to prevent any undesirable alternation in nucleic acid substrates by enzymatic activities of integrase proteins, such as 3'-processing. Notably, nucleic acid substrate consisted of an R-loop structure bound to HIV-1 integrase proteins with higher binding affinity than that of the simple dsDNA duplex (**Fig. 5A** [↗](#)). Additionally, the R-loop composing forms of nucleic acid structures, such as RNA-DNA hybrid with exposed ssDNA (R:D+ssDNA) and RNA-DNA hybrid (hybrid), also showed high binding affinity to integrases (S6A Fig. and **Fig. 5A** [↗](#)).

We validated the interaction between cellular genomic R-loops and HIV-1 integrase proteins by DNA-RNA hybrid immunoprecipitation using S9.6 antibodies against FLAG-tagged HIV-1 integrase-expressing HeLa cells (**Fig. 5B** [↗](#)). Under our experimental conditions, R-loops were reproducibly immunoprecipitated (S6B Fig.) and HIV-1 integrase proteins co-immunoprecipitated with the R-loops (**Fig. 5C** [↗](#)). DNA-RNA hybrids also co-immunoprecipitated with the positive control H3 ([39](#) [↗](#)) but not with the negative controls LaminA/C and Actin ([39](#) [↗](#)) (**Fig. 5C** [↗](#)). To verify the specificity of our co-immunoprecipitation results for R-loops and HIV-1 integrases, we performed DNA-RNA hybrid immunoprecipitation with RNase H treatment (S6C Fig.). The S9.6 signal of the

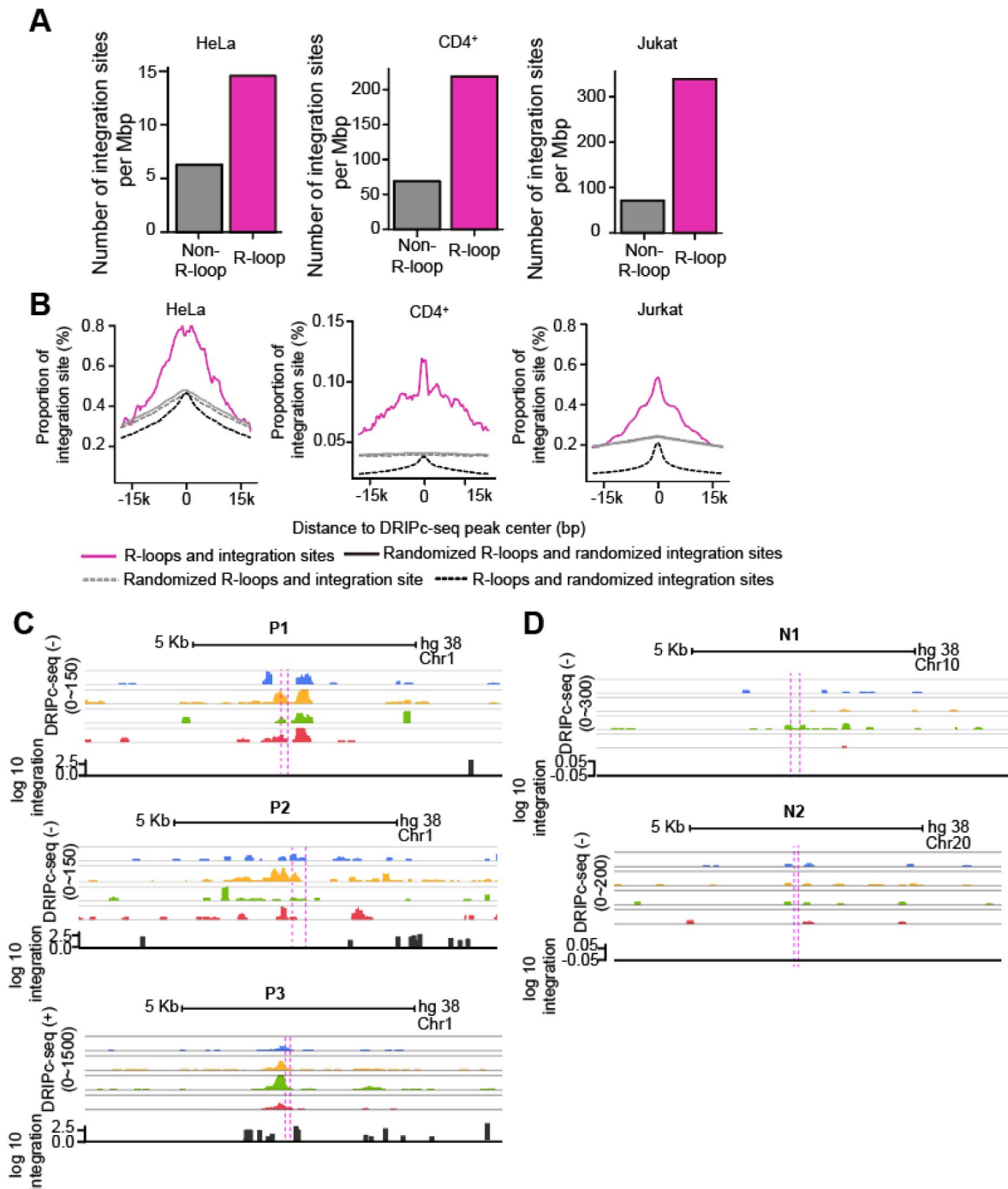


Fig. 4.

HIV-1 prefers host genomic R-loop regions for its viral cDNA integration.

(A) Bar graphs showing quantified number of HIV-1 integration sites per Mb pairs in total regions of 30-kb windows centered on DRIPc-seq peaks from HIV-1 infected HeLa cells, primary CD4⁺ T cells and Jurkat cells (magenta) or non-R-loop region in the cellular genome (gray). (B) Proportion of integration sites within the 30-kb windows centered on DRIPc-seq peaks (magenta solid lines) or randomized DRIPc-seq peaks (gray dotted lines). Control comparisons between randomized integration sites with DRIPc-seq peaks and randomized DRIPc-seq peaks are indicated by black dotted lines and gray solid lines, respectively. (C and D) Superimpositions of HIV-1-induced R-loop positive chromatin regions, P1-P3 (C), and HIV-1-induced R-loop negative chromatin regions, N1 and N2 (D), on DRIPc-seq (blue, 0 hpi; yellow, 3 hpi; green, 6 hpi; red, 12 hpi) and number of mapped read of HIV-1 integration-seq (integration, black). Magenta dotted lines represent primer binding sites in qPCR following DRIP.

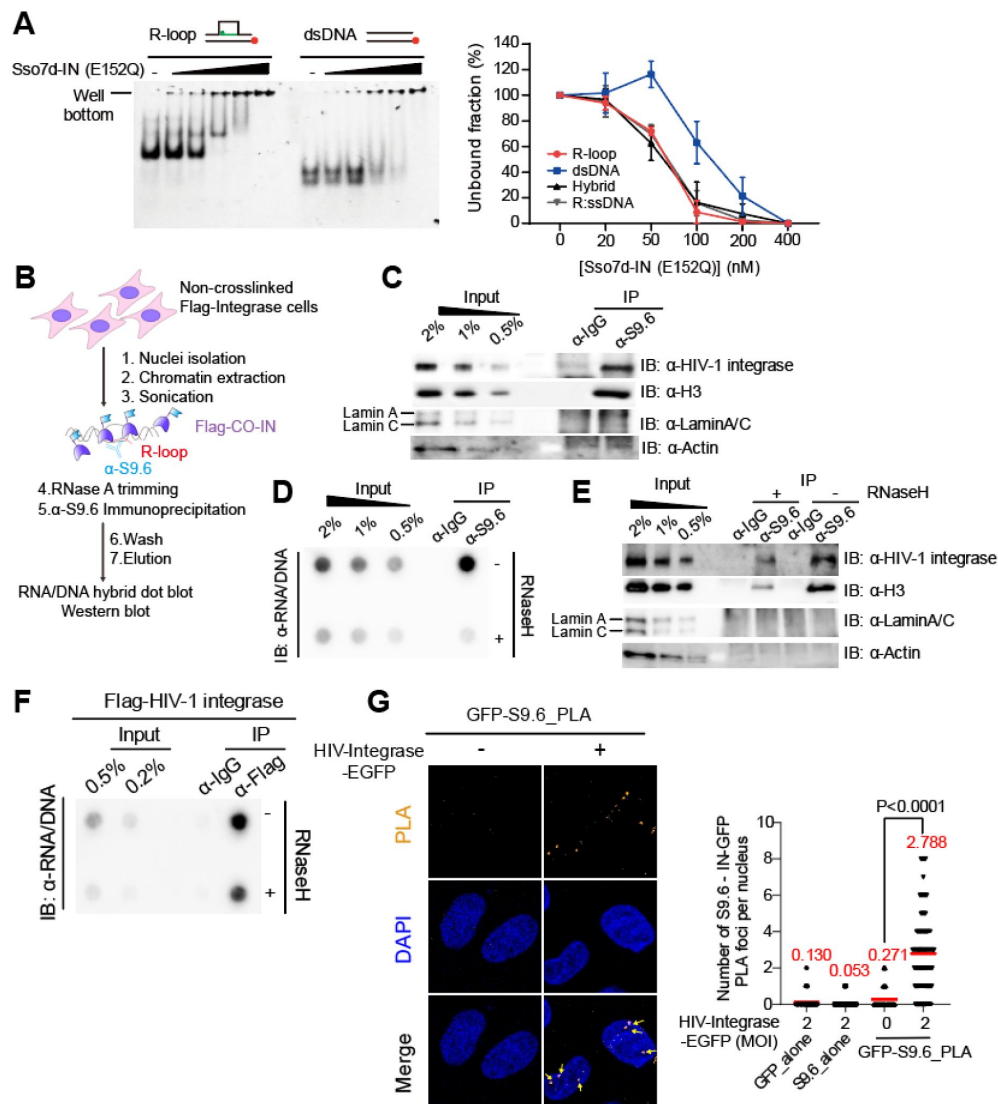


Fig. 5.

HIV-1 integrase proteins directly bind to host genomic R-loops.

(A) Representative gel images for EMSA of Sso7d-tagged HIV-1-integrase (E152Q) with R-loop and dsDNA, 10 nM nucleic acid substrate was incubated with Sso7d-tagged HIV-1-integrase (E152Q) at 0 nM, 20 nM, 50 nM, 100 nM, 200 nM, and 400 nM (left). Unbound fraction were quantified for EMSA of Sso7d-tagged HIV-1-integrase (E152Q) with different types substrates (R-loop, dsDNA, R-loop, R:D+ssDNA and Hybrid). Data are presented as the mean $\pm$ SEM, n = 3 (right). (B) Summary of the experimental design for R-loop immunoprecipitation using S9.6 antibody in FLAG-tagged HIV-1 integrase protein-expressing HeLa cells. (C) Western blotting for HIV-1 integrase protein, H3, and LaminA/C of DNA-RNA hybrid immunoprecipitation using the S9.6 antibody. (D) and (E) HeLa gDNA input was either untreated (-) or treated (+) with RNase H before enrichment for DNA-RNA hybrids using the S9.6 antibody. gDNA-RNA hybrids were incubated with nuclear extracts depleted of DNA-RNA hybrids with RNase A followed by S9.6 immunoprecipitation. DNA-RNA hybrid dot blot (D) and western blot of DNA-RNA hybrid immunoprecipitation, probed with the indicated antibodies (E). (F) DNA-RNA hybrid dot blot of FLAG antibody-immunoprecipitated nucleic acid extracts. Where indicated, nucleic acid extracts were untreated (-) or treated (+) with RNase H before probing with the S9.6 antibodies. (G) Representative images of the proximity-ligation assay (PLA) between GFP and S9.6 antibodies in HIV-IN-EGFP virion-infected HeLa cells at 6 hpi. Cells were subjected to PLA (orange) and co-stained with DAPI (blue). PLA puncta in the nucleus are indicated by the yellow arrows. Quantification analysis of number of PLA foci per nucleus (left). GFP_alone and S9.6_alone were used as single-antibody controls from HIV-IN-EGFP virion-infected HeLa cells (right). The mean value for each data point is indicated by the red line. P value was calculated using a two-tailed unpaired t-test (n > 50).

immunoprecipitated nucleic acids was highly sensitive to RNase H treatment of the pre-immunoprecipitates (**Fig. 5D**). Accordingly, the blotting signal of the co-immunoprecipitated HIV-1 integrase and H3 proteins was significantly reduced upon RNase H treatment (**Fig. 5E**). We performed reciprocal immunoprecipitation using an anti-FLAG monoclonal antibody and detected the immunoprecipitated R-loops using dot blot analysis with anti-S9.6. R-loops, which were immunoprecipitated using HIV-1 integrase, and the S9.6 signal of immunoprecipitated nucleic acids was highly sensitive to RNase H treatment (**Fig. 5F** and S6D Fig.). Subsequently, we examined the interaction between the R-loops and HIV-1 integrase using a proximity-ligation assay (PLA) in HIV-1-infected cells. We used two antibodies: one that binds to the R-loops (anti-S9.6) and another that binds to GFP-tagged HIV-1 integrase. We detected PLA signals in cells infected with HIV-IN-EGFP virions (34) and non-infected control cells. PLA signals in non-infected cells were comparable to those in S9.6-alone and GFP-alone single antibody-negative controls; however, PLA signals significantly increased upon HIV-1 infection (**Fig. 5G** and S6E Fig.). Our data suggest that the HIV-1 frequently targets R-loop-rich regions for viral genome integration by physically binding of HIV-1 integrase proteins to R-loop structures on the host genome.

Discussion

In this study, we found that HIV-1 preferentially integrated into regions rich in R-loops, suggesting that R-loops are novel host factors that contribute to HIV-1 integration site selection. In our bioinformatics analysis, host cellular R-loops were induced by VSV-G-pseudotyped HIV-1 infection and widespread in host genomic regions. Using our R-loop-inducible cell models, R-loop formation, and not necessarily the transcriptional activity itself, was found to be important for HIV-1 integration site determination. In addition, HIV-1 integrase proteins favor physical binding with R-loops *in vitro*, and interact with host genomic R-loops in HIV-1-infected cells. These results demonstrate that HIV-1 exploits and frequently targets the host genomic R-loop regions for successful integration and infection.

One possible explanation for why HIV-1 integration shows a preference for host genomic R-loop regions is that the R-loop structure may drive dynamics in the genomic environment and the spatial organization of the genome, resulting in increased accessibility of HIV-1 intasome binding to the target host genomic region. The R-loops display enhancer and insulator chromatin states, that can act as distal regulatory elements by recruiting diverse chromatin binding factors (22). This not only allows R-loops to drive dynamics in the genome, but also possibly drives R-loop-mediated integration over long-range genomic regions. R-loop regions exhibit increased chromatin accessibility. In the cellular genome, these structures relieve superhelical stresses and are often associated with open chromatin marks and active enhancers (22, 23), which are also distributed over HIV-1 integration sites (6). In the case of transcription-induced R-loop formation, a guanine-quadruplex (G4) structure can be generated in the non-template DNA strand of the R-loop, which is another contributor to genome architecture (24). A recent study showed that G4 DNA can influence both productive and latent HIV-1 integration, as well as the potential for reactivation of latent proviruses (25).

Another possible explanation for the preference of HIV-1 integration for host genomic R-loops is that R-loops may play a collaborative role with host factors governing the HIV-1 integration site selection. Cellular R-loops are recognized and regulated by numerous cellular proteins (39, 40). LEDGF/p75 (9, 13, 41) and CPSF6 (7, 9) are two decisive host factors that direct HIV-1 integration by interacting with integrase or trafficking the viral preintegration complex towards the nuclear interior (7, 9). In fact, these host factors have recently been identified as potential R-loop-binding proteins in DNA–RNA interactome analysis (39) and R-loop proximity proteomics (40), respectively. R-loops are tightly regulated by DNA damage response proteins (42) and the DNA repair machinery plays an important role in HIV-1 integration process (32). For example, the Fanconi anemia pathway (43, 44), a well-known R-loop regulatory pathway,

has been recently proposed as an HIV-1 integration regulatory factor exploited by HIV-1 (45 [↗](#)). Considering these previous studies and our current findings, we propose R-loops as another putative host factor driving HIV-1 integration site determination, and a possible intermediate regulator of HIV-1 integration site selection by such host proteins.

Our data showed that HIV-1 integrase proteins physically interacted with genomic R-loops *in vitro* and in cells. Recent advancements in cryogenic electron microscopy (cryo-EM) technology have revealed the conformational characteristics of the target DNA during retroviral integration (27 [↗](#), 28 [↗](#)). During retroviral integration, the target DNA undergoes a transition of its conformation from B-form to A-form. R-loops, which represent intermediates between the B-form DNA and A-form RNA conformations (27 [↗](#)), may have an intrinsic preferential binding ability to retroviral intasomes over other nucleic acid structures.

Viruses often take advantage of various host factors, and targeting the viral components that manipulate the host cellular environment can be an effective strategy for antiviral therapy. Our study showed the host genomic R-loops significantly accumulate shortly after HIV-1 infection. Thus, it is possible that virion-associated HIV-1 proteins are responsible for inducing these R-loops. For instance, the HIV-1 accessory protein Vpr causes genomic damage (46 [↗](#)) and transcriptomic changes during the early stages post infection (47 [↗](#)), both of which can lead to *in cis* and *in trans* R-loop formation (16 [↗](#)). Another HIV-1 accessory protein, Vif, counteracts the host antiviral factor, APOBEC3 (48 [↗](#), 49 [↗](#)), which was recently found to regulate cellular R-loop levels (50 [↗](#)). Identifying the HIV-1 components responsible for inducing host cellular R-loops, elucidating the mechanism by which they induce genome-wide R-loop formation, and contributing to successful viral integration into selective genomic regions, are areas for further research.

Although most HIV-1 integration occurs in genic regions (4 [↗](#), 6 [↗](#)), HIV-1 proviruses are also found in non-genic regions (51 [↗](#)) and understanding these “transcriptionally silent” proviruses is critical for developing strategies to completely eliminate HIV-1. In HIV-1 elite controllers, who suppress viral gene expression to undetectable levels, HIV-1 proviruses in heterochromatic regions are not eliminated but selected by the immune system (5 [↗](#)). Moreover, proviruses with lower expression level can persist in the host genome even during antiretroviral therapy (4 [↗](#)). However, the mechanism by which HIV-1 targets gene-silent regions for “invisible” integration remains unclear. Our study has revealed that R-loops are enriched in both genic and non-genic regions during HIV-1 infection, and that the virus preferentially targets these R-loops for integration. We propose that R-loops, particularly those enriched in non-genic regions, may represent the mechanism by which the virus achieves “invisible” and permanent infection.

Materials and methods

Cell culture

HeLa and HEK293T cells were cultured in Dulbecco's modified Eagle's medium (Gibco) supplemented with 10% (v/v) fetal bovine serum (FBS, Cytiva), antibiotic mixture (100 units/ml penicillin–streptomycin, Gibco), and 1% (v/v) GlutaMAX-I (Gibco). Jurkat cells were cultured in Roswell Park Memorial Institute (RPMI) 1640 medium (ATCC) supplemented with 10% (v/v) FBS (Cytiva). Cells were incubated at 37°C and 5% CO₂.

Virus production and infection

VSV-G-pseudotyped HIV-1 virus stocks were prepared by performing standard polyethylenimine-mediated transfection of HEK293T monolayers with pNL4-3 ΔEnv EGFP (NIH AIDS Reagent Program 11100) or pNL4-3. Luc.R-E (NIH AIDS Reagent Program, 3418) along with pVSV-G at a ratio of 5:1. HIV-IN-EGFP virions were produced as previously described (34 [↗](#)) by performing polyethylenimine-mediated transfection of HEK293T cells with 6 μg of pVpr-IN-EGFP, 6 μg of HIV-1

NL4-3 non-infectious molecular clone (pD64E; NIH AIDS Reagent Program 10180), and 1 μ g of pVSV-G. The cells were incubated for 4 h before the medium was replaced with fresh complete medium. Virion-containing supernatants were collected after 48 h, filtered through a 0.45- μ m syringe filter, and pelleted using the Lenti-X Concentrator (631232; Clontech) according to the manufacturer's instructions. The multiplicity of infection (MOI) of virus stocks was determined by transducing a known number of HeLa cells with a known amount of virus particles and then counting GFP-positive cells using flow cytometry. For luciferase reporter HIV-1 virus, the HIV-1 p24 antigen content in viral stock were quantified using the HIV1 p24 ELISA kit (Abcam, ab218268), according to the manufacturer's instruction. For virus infection, HeLa cells were seeded at a density of $0.5\text{--}4 \times 10^5$ cells/mL on the day before infection. The culture medium was replaced with fresh complete culture medium 2 hpi. The infected cells were washed twice with PBS and harvested at the indicated time points. Jurkat cells were seeded at a density of 1×10^6 cells/mL and inoculated with 300ng/p24 capsid antigen. The plates were centrifuged at 1000 g at 30°C for 1 h. The medium was replaced with fresh RPMI 2 h after infection.

Primary cell isolation, culture, T cell activation, and infection

For CD4⁺ T cells isolation, human PBMC (ST70025, STEMCELL Technologies) was mixed and incubated with MACS CD4 MicroBeads (130-045-101, Miltenyi Biotec) and FITC-conjugated mouse anti-CD4 (561005, BD Bioscience) according to the manufacturer's instructions. Then the CD4⁺ T cells were enriched by using LS Columns (130-042-401, Miltenyi Biotec) and MidiMACS Separator (130-042-302, Miltenyi Biotec). The efficiency of magnetic separation was analyzed by using Flow-Activated Cell Sorter Canto II (BD Bioscience) and Flowjo software (Flowjo).

CD4⁺ T cells were cultured in Roswell Park Memorial Institute (RPMI) 1640 medium (Gibco), supplemented with 10% (v/v) fetal bovine serum (FBS, Cytiva), antibiotic mixture (100 units/ml penicillin–streptomycin, Gibco), 1% (v/v) GlutaMAX-I (Gibco), and 20 ng/ml of IL-2 (PHC0026, Gibco), left in resting state or activated with Dynabeads Human T-Activator CD3/CD28 (1161D, Thermo Fisher Scientific) for 72 h. CD4⁺ T cells activation efficiency was assessed by staining cells with FITC-conjugated mouse anti-CD25 (340694, BD Bioscience) and APC-conjugated mouse anti-CD69 (130-114-046, Miltenyi Biotec) and using Flow-Activated Cell Sorter Canto II (BD Bioscience) and Flowjo software (Flowjo).

Purified and activated CD4⁺ T cells were seeded at a density of 1×10^6 cells/mL and inoculated with 600ng/p24 capsid antigen in presence of polybrene. The plates were centrifuged at 1000 g at 30°C for 1 h. The medium was replaced with fresh RPMI 2 h after infection.

DRIP-qPCR

DRIP was performed as described for the construction of the DRIPc-seq library. After the elution of isolated complexes, nucleic acids were purified using the standard phenol-chloroform extract method and used for qPCR. S6 Table presents details of the primer sequences used for DRIP-qPCR analysis.

RNA-seq library construction

For RNA-seq, HeLa cells were infected with VSV-G-pseudotyped HIV-1 NL4-3 Δ Env EGFP virus at a MOI of 0.6 and harvested at 0, 3, 6, and 12 hpi. Sequencing was performed with biological replicates. Total RNA was extracted using TRIzol reagent (Invitrogen), according to the manufacturer's instructions. An mRNA sequencing library was constructed using Illumina adaptors harboring p5 and p7 sequences and Rd1 SP and Rd2 SP sequences. Sequencing was performed using the HiSeq2500 system (Illumina).

Luciferase assay

HeLa cells infected with VSV-G-pseudotyped pNL4-3.Luc.R-E HIV-1 viruses were harvested at 48 hpi, and luminescence was measured using the Dual-Luciferase Reporter Assay System (Promega) according to the manufacturer's instructions. Briefly, 250 μ l of passive lysis buffer was used to lyse cells for each sample, 20 μ l of the lysate was mixed with 100 μ l of the Luciferase Assay Reagent II, and the luminescence of firefly luciferase was measured using a microplate luminometer (Berthold). The luminescence signal were normalized with total protein content, measured by BCA assay.

Quantitative real-time PCR (qPCR)

For RT (reverse transcription)-qPCR, 1 μ g of RNA was reverse-transcribed using the ReverTra Ace qPCR RT Kit (TOYOBO) following the manufacturer's instructions. For qPCR, DNA extracts were prepared using a DNA purification kit (Qiagen, 51106) according to the manufacturer's instructions. Equivalent amounts of purified gDNA from each sample were analyzed using qPCR. qPCR was performed using TOPreal qPCR PreMIX (Enzynomics, RT500M). The reactions were performed in duplicate or triplicate for technical replicates. PCR was performed using the iCycler iQ real-time PCR detection system (Bio-Rad). All the primers used for qPCR are listed in S6 Table.

DRIPc-seq library construction

DRIP followed by library preparation, next-generation sequencing, and peak calling were performed as described earlier ([29](#)). Briefly, the corresponding cells were harvested and their gDNA was extracted. The extracted nucleic acids were fragmented using a restriction enzyme cocktail with BsrB I (NEB, R0102S), HindIII (NEB, R0136L), Xba I (NEB, R0145L), and EcoRI (NEB, R3101L) overnight at 37°C. Half of the fragmented nucleic acids were digested with RNase H (New England Biolabs) overnight at 37°C to serve as a negative control. The digested nucleic acids were cleaned using standard phenol-chloroform extraction and resuspended in DNase/RNase-free water. DNA–RNA hybrids were immunoprecipitated from total nucleic acids using mouse anti-DNA–RNA hybrid S9.6 (Kerafast, ENH001) DRIP binding buffer and incubated overnight at 4°C. Dynabeads Protein A (Invitrogen, 10001D) was used to pull down the DNA-antibody complexes by incubation for 4 h at 4°C. The isolated complexes were washed twice with DRIP binding buffer before elution. For elution, the isolated complexes were incubated in an elution buffer containing proteinase K for 45 min at 55 °C. Subsequently, DNA was purified using the standard phenol-chloroform extract method and subjected to DNase I (Takara, 2270 B) treatment and reverse transcription for DRIPc-seq library construction. DRIPc-seq was performed in biological replicates. S5 Table shows details of the oligonucleotides used for DRIPc-seq library construction. DRIPc-seq libraries were analyzed using 150 bp paired-end sequencing on a HiSeqX Illumina instrument.

Immunofluorescence microscopy

For immunofluorescence assays of S9.6 nuclear signals, when indicated, the cells were pre-extracted with cold 0.5% NP-40 for 3 min on ice. Cells were fixed with 100% ice-cold methanol for 10 min on ice and then incubated with 100% ice-cold acetone for 1 min. The slides were washed three times with 1 $\times$ PBS and incubated with or without 60 U/mL RNase H (M0297S, NEB) at 37°C for 36 h or left untreated. The slides were subsequently briefly rinsed thrice with 2% BSA/0.05% Tween (in PBS) and incubated with mouse anti-DNA–RNA hybrid S9.6 (Kerafast, ENH001; 1:100) and rabbit anti-nucleolin (Abcam, ab22758; 1:300) in 2% BSA/0.05% Tween (in PBS) for 4 h at 4°C. The slides were then washed three times with 2% BSA/0.05% Tween (in PBS) and incubated with goat anti-rabbit AlexaFluor-488-conjugated (Invitrogen, A-11008) and goat anti-mouse AlexaFluor-568-conjugated (Molecular Probes, A11004) secondary antibodies (1:200) for 2 h at room temperature. The slides were then washed three times with 2% BSA/0.05% Tween (in PBS) and mounted using the ProLong Gold AntiFade reagent (Invitrogen). Images were obtained using an inverted microscope Nikon Eclipse Ti2, equipped with a 1.45 numerical aperture, plan apochromat

lambda 100× oil objective, and an scientific complementary metal–oxide–semiconductor camera (Photometrics prime 95 B 25 mm). For each field of view, images were obtained with DAPI395, GFP488, and Alexa594 channels using the NIS-Elements software. For quantification analysis, binary masks of nuclei and nucleoli were generated using the ROI manager and auto local thresholding using the ImageJ software. The intensity of nuclear signals for DNA–RNA hybrids and nucleolin was then quantified. The final DNA–RNA hybrid signals in the nucleus were calculated by subtracting the nucleolin signals from the DNA–RNA hybrid signals.

pgR-rich and –poor cell line generation with piggyBac transposition

We adapted and modified an elegantly designed episomal system that induces defined R-loops with controlled transcription levels ([17](#)) for R-loop-forming or non-R-loop-forming sequence subcloning into the piggyBac transposon vector. HeLa cells were seeded at a density of 5×10^4 cells/ml in a 6-well plate. The next day, cells were transfected with 0.2 µg of Super PiggyBac Transposase Expression Vector (System Biosciences, PB210PA-1) and 0.2, 1, or 2 µg of transposon vectors with appropriate “cargo” sub cloned using Lipofectamine 3000 (Invitrogen) according to the manufacturer’s instructions. After 3 days, the cells were treated with 10 µg/ml blasticidin S (Gibco, A1113903) for selection. Cells with positive integrants for more than 7 days were validated using immunoblotting or RT-qPCR following treatment with DOX. Jurkat cells were seeded at a density of 8×10^5 cells/ml in a 6-well plate and transfected with 0.2 µg of transposase and 1 µg of corresponding transposon vectors with Lipofectamine 3000, like HeLa cells. After 3 days, the cells were treated with 10 µg/ml blasticidin S (Gibco, A1113903) for selection. For each passage, cells were cushioned onto Ficoll-Pacque (Cytiva, 17144002) to separate live cells from dead cell debris. The cells over the cushion were washed with PBS and incubated in cell culture medium with 10 µg/ml of blasticidin for further selection for at least 14 days. Cells with positive integrants were validated by immunoblotting after treatment with DOX. Quantification of successfully integrated piggyBac transposons was performed using a piggyBac qPCR copy number kit (System Biosciences, PBC100A-1).

HIV-1 integration site sequencing library construction

HIV-1 integration site sequencing library construction was performed as described earlier ([7](#), [9](#)). Summarily, HeLa cells and primary CD4⁺ T cells were infected with VSV-G-pseudotyped HIV-1 NL4-3 ΔEnv EGFP virus at a MOI of 0.6 and harvested 5 days post infection. gDNA was isolated using a DNA purification kit (Qiagen, 51106), according to the manufacturer’s instructions. gDNA (10 µg) was digested overnight at 37°C with 100 U each of the restriction endonucleases MseI (NEB, R0525L) and BglII (NEB, R0144L). Linker oligonucleotides, which were compatible for ligation with the MseI-generated DNA ends, were ligated with gDNA overnight at 12°C in reactions containing 1.5 µM ligated linker, 1 µg fragmented DNA, and 800 U T4 DNA ligase (NEB, M0202S). Viral LTR–host DNA junctions were amplified using semi-nested PCR with a unique linker-specific primer and LTR primers. The second round of PCR was carried out with primers binding to the LTR and the linkers for next-generation sequencing. Two PCRs were performed in parallel for the first round of PCR and five PCRs were performed in parallel for the second round of PCR to enhance library diversity. S7 Table presents details of the oligonucleotides used for HIV-1 integration site sequencing library construction. HIV-1 integration site sequencing was performed in biological replicates. Integration site libraries were analyzed using 150 bp paired-end sequencing on a HiSeqX Illumina instrument.

Co-immunoprecipitation of DNA–RNA hybrid

DNA–RNA hybrid immunoprecipitation was performed as described earlier ([39](#)). Summarily, non-crosslinked HeLa cells transfected with the pFlag-IN codon-optimized plasmid were lysed in 85 mM KCl, 5 mM PIPES (pH 8.0), and 0.5% NP-40 for 10 min on ice, and then, the lysates were centrifuged at 750 *g* for 5 min to pellet the nuclei. The pelleted nuclei were resuspended in sodium

deoxycholate, SDS, and sodium lauroyl sarcosinate in RSB buffer and were sonicated for 10 min (Diagenode Bioruptor). Extracts were then diluted (1:4 in RSB + T buffer) and subjected to immunoprecipitation with the S9.6 antibody overnight at 4°C. Antibody-bound complexes were incubated with Protein A Dynabeads (Invitrogen) for 4 h at 4°C for immunoprecipitation. Normal mouse IgG antibodies (Santa Cruz, sc-2025) were used as negative controls. RNase A (Thermo Scientific, EN0531) was added during immunoprecipitation at 0.1 ng RNase A per µg gDNA. Beads were washed four times with RSB + T; twice with RSB, and eluted either in 2× LDS (Novex, NP0007), 100 mM DTT for 10 min at 70°C (for western blot), or 1% SDS and 0.1 M NaHCO₃ for 30 min at room temperature (for DNA–RNA hybrid dot blot).

For co-immunoprecipitation of DNA–RNA hybrids with RNase H treatment, gDNA containing RNA–DNA hybrids was isolated from HeLa cells transfected with a pFlag-IN codon-optimized plasmid using a QIAmp DNA Mini Kit (Qiagen, 51304). gDNA was sonicated for 10 min (Diagenode Bioruptor) and then treated with 5.5 U RNase H (NEB, M0297) per µg of DNA overnight at 37 °C. A fraction of gDNA was stored as “nucleic acid input” for dot blot analysis. gDNA was cleaned using standard phenol-chloroform extraction, resuspended in DNase/RNase-free water, enriched for DNA–RNA hybrids using immunoprecipitation with the S9.6 antibody (overnight at 4°C), isolated with Protein A Dynabeads (Invitrogen; 4 h at 4°C), washed thrice with RSB+T. The immunoprecipitated complexes were incubated with nuclear extracts of HeLa cells transfected with the pFlag-IN codon-optimized plasmid for 2 h at 4°C with diluted HeLa nuclear extracts. The cell lysate containing proteins were pre-treated with 0.1 mg/ml RNase A (Thermo Scientific, EN0531) for 1 h at 37°C to degrade all RNA–DNA hybrids, and the excess of RNase A was blocked by adding 200 U of SUPERase in RNase inhibitor (Invitrogen, AM2694) for immunoprecipitation. In addition, 100 µL fraction of diluted and RNase A pre-treated extracts prior to immunoprecipitation was stored as “protein input” for western blotting. Beads were washed four times with RSB + T; twice with RSB, and eluted either in 2× LDS (Novex, NP0007), 100 mM DTT for 10 min at 70°C (for western blot), or 1% SDS, and 0.1 M NaHCO₃ for 30 min at room temperature (for DNA–RNA hybrid dot blot).

Recombinant Sso7d-IN protein purification

Sso7d-integrase active site mutant E152Q was expressed in *Escherichia coli* BL21-AI and purified essentially as previously described (52 [↗](#)). Briefly, Sso7d-IN (E152Q) expressed BL21-AI cells were lysis in lysis buffer (20 mM HEPES pH 7.5, 2 mM 2-mercaptoethanol, 1 M NaCl, 10% (w/v) glycerol, 20 mM imidazole, 1 mg RNase A, and 1000 U DNase I) and purified by nickel affinity chromatography (Qiagen, 30210). Protein were first loaded on HeparinHP column (GE Healthcare) equilibrated with equilibrated with 20 mM Tris, pH 8.0, 0.5 mM TCEP, 200 mM NaCl, 10% glycerol for anion exchange chromatography prior to the size exclusion chromatography. Proteins were eluted with a linear gradient of NaCl from 200 mM to 1 M. Eluted fractions were pooled and then separated on a Superdex-200 PC 10/300 GL column (GE Healthcare) equilibrated with 20 mM Tris pH 8.0, 0.5 mM TCEP, 500 mM NaCl and 6% (w/v) glycerol. The purified protein was concentrated to 0.6 mg/ml using an Amicon centrifugal concentrator (EMD Millipore), flash-frozen in liquid nitrogen and stored at –80°C.

Electrophoretic mobility shift assay for R-loop binding of Sso7d-IN

To test the binding affinity of Sso7d-tagged HIV-1 IN to different types of nucleic acid substrates, we prepared R-loop, dsDNA, RNA–DNA hybrid with exposed ssDNA (R:D+ssDNA), RNA–DNA hybrid (Hybrid), ssDNA, and ssRNA by annealing different combinations of Cy3, Cy5 or non-labeled oligonucleotides following the previous protocol (53 [↗](#)). 10 nM of DNA substrate was incubated with Sso7d-IN at different concentrations in assembly buffer (20 mM HEPES pH 7.5, 5 mM CaCl₂, 8 mM 2-mercaptoethanol, 4 uM ZnCl₂, 100 mM NaCl, 25% (w/v) glycerol and 50 mM 3-(Benzyltrimethylammonio) propanesulfonate (NDSB-256)), for 1 h at 30°C then incubated for 15

min on ice. All the reactants were run on 4.5% non-denaturing PAGE in 1× TBE and then Cy3 or Cy5 fluorescence signal was imaged by ChemiDoc MP imaging system (Bio-Rad). S8 Table presents details of the oligonucleotide sequence used for EMSA.

PLA

For PLA, HeLa cells were grown on coverslips and infected with HIV-IN-EGFP virions. At 6 hpi, cells were pre-extracted with cold 0.5% NP-40 for 3 min on ice. The cells were fixed with 4% paraformaldehyde in PBS for 15 min at 4 °C. The cells were then blocked with 1× blocking solution (Merck, DUO92102) for 1 h at 37°C in a humidity chamber. After blocking, cells were incubated with the following primary antibodies overnight at 4°C for S9.6-HIV-1-IN_PLA: mouse anti-DNA–RNA hybrid S9.6 (1:250; Kerafast, ENH001) and rabbit anti-GFP (1:500; Abcam, ab6556). The following day, after washing with once with buffer A twice (Merck, DUO92102), cells were incubated with pre-mixed Duolink PLA plus (anti-mouse) and PLA minus probes (anti-rabbit) antibodies for 1 h at 37°C. The subsequent steps in the proximal ligation assay were performed using the Duolink PLA Fluorescence kit (Sigma) according to the manufacturer’s instructions. To obtain images, the mounted specimens were visually scanned and representative images were acquired using a Zeiss LSM 710 laser scanning confocal microscope (Carl Zeiss). The number of intranuclear PLA puncta was quantified using the ImageJ software. For each biological replicate and experiment, a PLA with a single antibody was performed as a negative control under the same conditions.

DRIPc-Seq data processing and peak calling

DRIPc-seq reads were quality-controlled using FastQC v0.11.9 ([54](#)), and sequencing adapters were trimmed using Trim Galore! v0.6.6 ([55](#)) based on Cutadapt v2.8 ([56](#)). Trimmed reads were aligned to the hg38 reference genome using bwa v0.7.17-r1188 ([57](#)). Read deduplication and peak calling were performed using MACS v2.2.7.1 ([58](#)). Because R-loops appear as both narrow and broad peaks in DRIPc-seq read alignment owing to its variable length, two independent “MACS2 callpeak” runs were performed for narrow and broad peak calling. The narrow and broad peaks were merged using Bedtools v2.26.0 ([59](#)). To increase the sensitivity of DRIPc-seq peak identification, peaks were called after pooling the two biological replicates of the DRIPc-seq sequencing data for each condition.

Consensus R-loop peak calling

The R-loop peaks at 0, 3, 6 and 12 hpi were first merged using “bedtools merge” to create a universal set of R-loop peaks across time points (n = 46542). Then, each of the universal R-loop peaks was tested for overlap with the R-loop peaks for 0, 3, 6 and 12 hpi using “bedtools intersect”. In all, 9,190, 21,403, 33,544, and 9,941 peaks overlapped with 0, 3, 6, and 12 hpi R-loop peaks, respectively. For CD4 cells, we identified a universal R-loop set consisting of 3,928 R-loops, and among them, 737, 722, 1,796 and 2,766 peaks overlapped with 0, 3, 6 and 12hpi R-loop peaks.

HIV-1 integration site sequencing data processing

Quality control of HIV-1 integration site-sequencing reads was performed using FastQC v0.11.9. To discard primers and linkers specific for integration site-sequencing from reads, we used Cutadapt v2.8 with the following option: “-u 49 -U 38 —minimum-length 36 —pair-filter any —action trim —q0,0 —a linker —A TGCTAGAGATTTCCACACTGACTGGGTCTGAGGG —A GGGTCTGAGGG —no-indels —overlap 12”. This allowed the first position of the read alignment to directly represent the genomic position of HIV-1 integration. Processed reads were aligned to the hg38 reference genome using bwa v0.7.17-r1188, and integration sites were identified using an in-house Python script. Genomic positions supported by more than five read alignments were regarded as HIV-1 integration sites. For Jurkat cells, we adopted integration site sequencing data of HIV-1 infected wild type Jurkat cells from SRR12322252 ([60](#)).

Co-localization analysis of R-loops and integration sites

Enrichment of integration sites near the R-loop peaks was tested using a randomized permutation test. Randomized R-loop peaks were generated using “bedtools shuffle” command, thus preserving the number and the length distribution of the R-loop peaks during the randomization process. Similarly, integration sites were randomized using the “bedtools shuffle” command. Randomization was performed 100 times. ENCODE blacklist regions ([61](#)) were excluded while shuffling the R-loops and integration sites to exclude inaccessible genomic regions from the analysis. For each of the observed (or randomized) integration sites, the closest observed (or randomized) R-loop peak and the corresponding genomic distance were identified using the “bedtools closest” command. The distribution of the genomic distances was displayed to show the local enrichment of integration sites in terms of the increased proportion of integration sites within the 30-kb window centered on R-loops compared to their randomized counterparts.

DNA plasmid construction and transfection

R-loop-forming mAIRN and non-R-loop forming ECPF sequences were subcloned from pSH26 and pSH36 plasmids, which were generously provided by Prof. Karlene A. Cimprich, into the piggyBac transposon vector, where the tet operator sequences were located upstream of the minimal CMV promoter. The pFlag-IN codon-optimized plasmid and pVpr-IN-EGFP were kindly provided by Prof. A. Engelman and Prof. Anna Cereseto, respectively. Lipofectamine 3000 (Invitrogen) transfection reagent was used for the transfection of all plasmids into cells, according to the manufacturer’s protocol.

DNA–RNA hybrid dot blotting

Total gDNA was extracted using the QIAmp DNA Mini Kit (Qiagen, 51304) according to the manufacturer’s instructions. gDNA (1.2 µg) was treated with 2 U RNase H (NEB, M2097) per µg of gDNA for 4 h at 37°C, with half of the sample left untreated but denatured. Half of the DNA sample was probed with S9.6 antibody (1:1000), and the other half was probed with an anti-ssDNA antibody (MAB3034, Millipore, 1:10000).

Immunoblotting

Cells were lysed using RIPA buffer (50 mM Tris, 150 mM sodium chloride, 0.5% sodium deoxycholate, 0.1% SDS, and 1.0% NP-40) supplemented with 10 µM leupeptin (Sigma-Aldrich) and 1 mM phenylmethanesulfonyl fluoride (Sigma-Aldrich) and boiled at 98°C for 10 min with SDS sample buffer prior to SDS-PAGE. The primary antibodies used were mouse monoclonal anti-FLAG M2 (Sigma, F3165), monoclonal mouse anti-HSC70 (Abcam, ab2788), polyclonal rabbit anti-histone H3 (tri methyl K4) antibody (Abcam, ab8580), monoclonal mouse anti-HIV-1 Integrase (Santa Cruz, sc-69721), rabbit anti-LaminA/C antibody (Cell Signaling, 2032), and monoclonal mouse anti-Actin (Invitrogen, MA1-744). All primary antibodies were used at a dilution of 1:1000 for western blotting. Peroxidase-conjugated anti-mouse IgG (115-035-062) and anti-rabbit IgG (111-035-003; both Jackson Laboratories) were used as secondary antibodies at 1:5000 dilution. Signals were detected using the SuperSignal West Pico chemiluminescence kit (Thermo Fisher Scientific).

RNA-seq data processing

RNA-seq reads were quality-controlled and adapter-trimmed as in DRIPc-seq processing. To quantify the expression levels of protein-coding genes, processed reads were aligned to the hg38 reference genome with GENCODE v37 gene annotation ([62](#)) using STAR v2.7.3a ([63](#)). Gene expression quantification was performed using RSEM v1.3.1. To quantify the expression levels of transposable elements (TEs), we used Tetranscripts v2.2.1 ([64](#)). Processed reads were first aligned to the hg38 reference genome using GENCODE v37 and RepeatMasker TE annotation using STAR v2.7.3a. In this case, STAR options were modified as follows to utilize multimapping reads in

downstream analyses: “--outFilterMultimapNmax 100 --winAnchorMultimapNmax 100 --outMultimapperOrder random --runRNGseed 77 --outSAMmultNmax 1 --outFilterType BySJout --alignSJoverhangMin 8 --alignSJDBoverhangMin 1 --alignIntronMin 20 --alignIntronMax 1000000 --alignMatesGapMax 1000000”. Expression levels of TEs were quantified as read counts with the “TEcount” command.

Genome annotations

All bioinformatic analyses were performed using the hg38 reference genome and GENCODE v37 gene annotation. Promoters were defined as a 2-kb region centered at the transcription start sites of the APPRIS principal isoform of protein-coding genes. TTS regions were defined as the 2-kb region centered at the 3' terminals of protein-coding transcripts. CpG island annotations were downloaded from the UCSC table browser. CpG shores were defined as 2-kb regions flanking CpG islands, excluding the regions overlapping with CpG islands. Similarly, CpG shelves were defined as 2-kb regions flanking the stretch of CpG islands and shores while excluding the regions overlapping with CpG islands and shores. Annotations for LINE, SINE, and LTR were extracted from the RepeatMasker track in the UCSC table browser.

Identification of viral sequencing reads in DRIPc-seq

To identify sequencing reads originating from the viral genome, we aligned DRIPc-seq reads to a composite reference genome consisting of the human and HIV1 genome (Genbank accession number: AF324493.2) and computed the proportion of the reads mapped to HIV1 genome.

Code availability

Bioinformatics pipelines and scripts used in this study are accessible from <https://github.com/dohlee/hiv1-rloop> .

Acknowledgements

We are grateful to Prof. Karlene A. Cimprich (Stanford University) for providing the pSH26 and pSH36 plasmids, Prof. A. Engelman (Harvard Medical School) for providing pFlag-IN codon optimized plasmid and Prof. Anna Cereseto (University of Trento) for providing pVpr-IN-EGFP. The NL4-3 ΔEnv EGFP and pNL4-3.Luc.R-E-viral plasmids were obtained through the NIH HIV Reagent Program, Division of AIDS, NIAID, NIH. We thank Dr. Sungchul Kim (IBS center for RNA Research) and Seongjin An (Korea University) for their technical support in recombinant protein purification.

Author contributions

K.P. and K.A. designed experiments. K.P., J.J. and S.L. performed experiments. D.L. performed the bioinformatical and statistical analyses. K.P., D.L., K.A. and S.K. analyzed the data. K.P., D.L., and K.A. wrote the manuscript.

Funding

This work was supported by the Institute for Basic Science of the Ministry of Science Grant (IBS-R008-D1) and the National Research Foundation of Korea (NRF) grant funded by the Korea government (NRF-2020R1A2C3011298) (to K. A.) and (NRF-2020R1A5A1018081) (to K.A.). The funders had no role in the study design, data collection, analysis, decision to publish, or preparation of the manuscript.

Competing interests

The authors have declared that no competing interests exist.

Supplemental Information

Materials and Methods

S1 Fig. Primary CD4⁺ T cells sorting strategies and GFP-HIV-1 infection

S2 Fig. Genome browser screenshot over the HIV-1-induced R-loop forming positive or negative genomic regions

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S1 Table. Chromosomal position and DRIPc-seq signal for referenced R-loop-positive and – negative regions in HIV-1 infected HeLa cells

S2 Table. Chromosomal position and DRIPc-seq signal for referenced R-loop-positive and – negative regions in HIV-1 infected primary CD4⁺ T cells

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S4 Table. RNA-seq analysis of relative gene expression levels of P1-3 and N1,2 R-loop regions

S5 Table. Oligonucleotides used for DRIPc-seq library construction

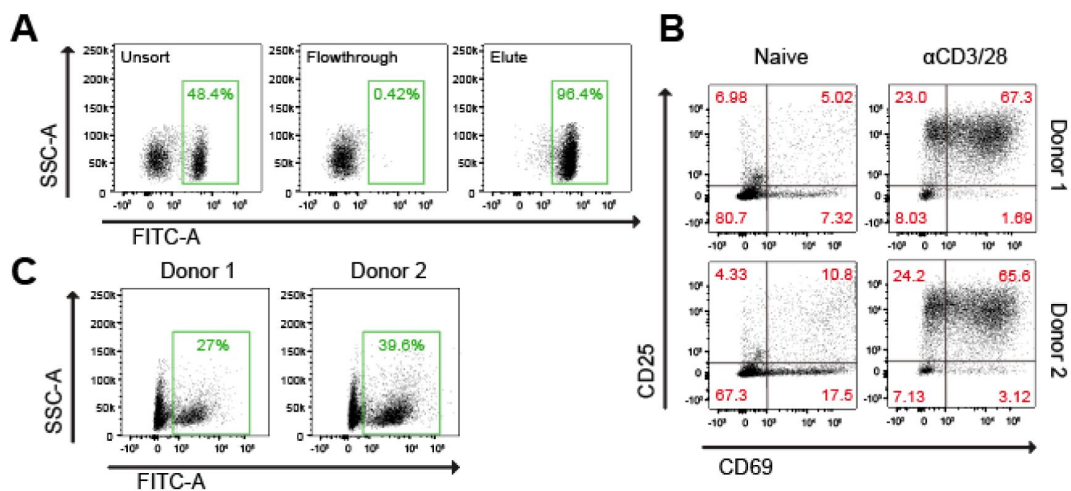
S6 Table. Primers used for qPCR

S7 Table. Oligonucleotides used for HIV-1 integration site sequencing library construct

S8 Table. Oligonucleotides used for electrophoretic mobility shift assay substrate preparation

S9 Table. Accession numbers and data sources.

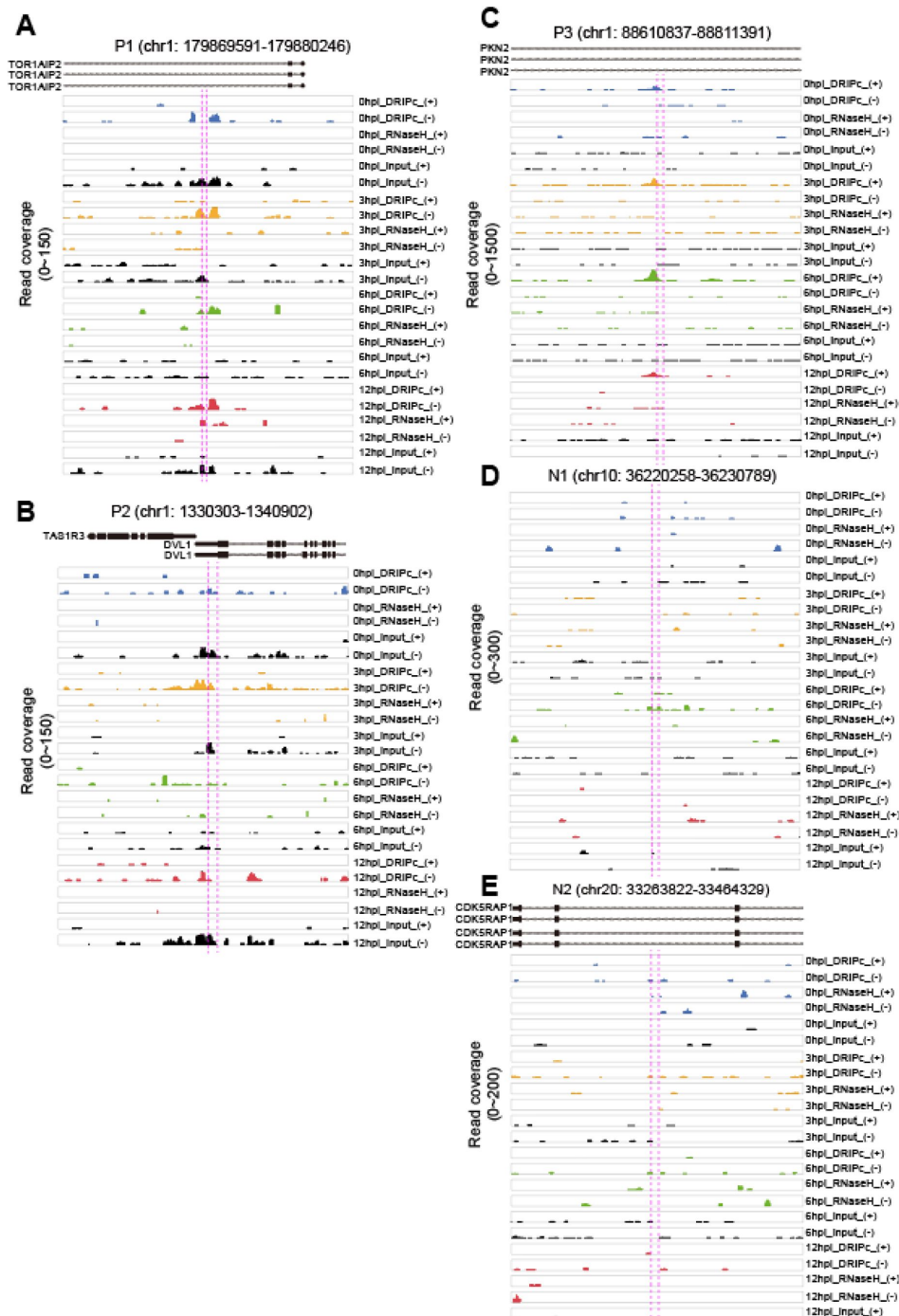
Supplementary figures



S1 Fig.

Primary CD4⁺ T cells sorting strategies and GFP-HIV-1 infection.

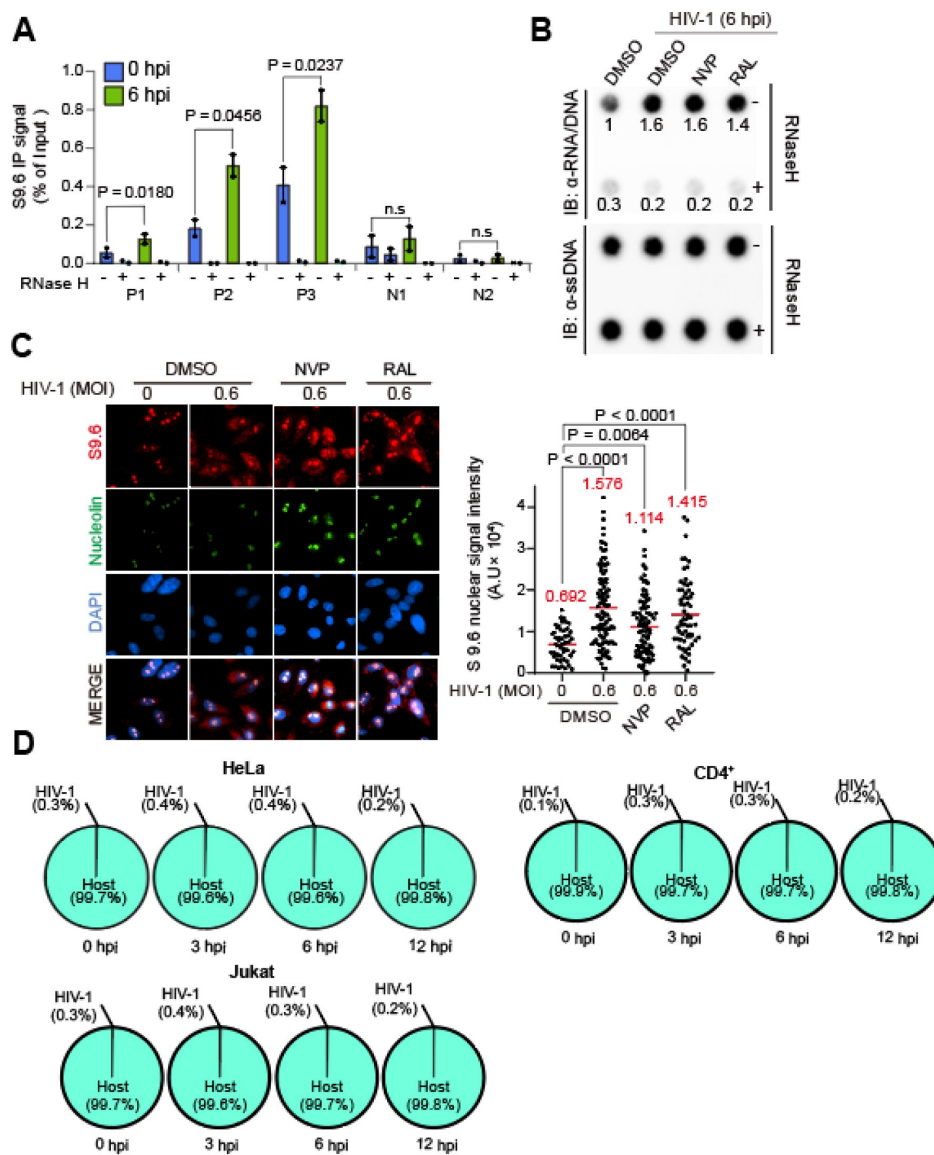
(A) Gating strategy used to determine the efficiency of CD4⁺ T cells sorting from human PBMC. Pre-sorted PBMCs were staining with FITC-conjugated anti-CD4 and subjected for positive CD4⁺ T cell sorting. The percentages of FITC stained cell population at each step of cell sorting are as indicated. (B) Gating strategy used to determine non-activated (Naïve) and activated cells (αCD3/28) with two markers, CD25 (FITC) and CD69 (APC), for each donor (upper panels, Donor 1; lower panels, Donor 2). (C) Gating strategy used to determine HIV-1-infectivity of CD4⁺ T cells from each donor infected with GFP reporter HIV-1 virus at 48 hpi. The percentages of GFP positive cell population at are as indicated.



S2 Fig.

Genome browser screenshot over the HIV-1-induced R-loop forming positive or negative genomic regions.

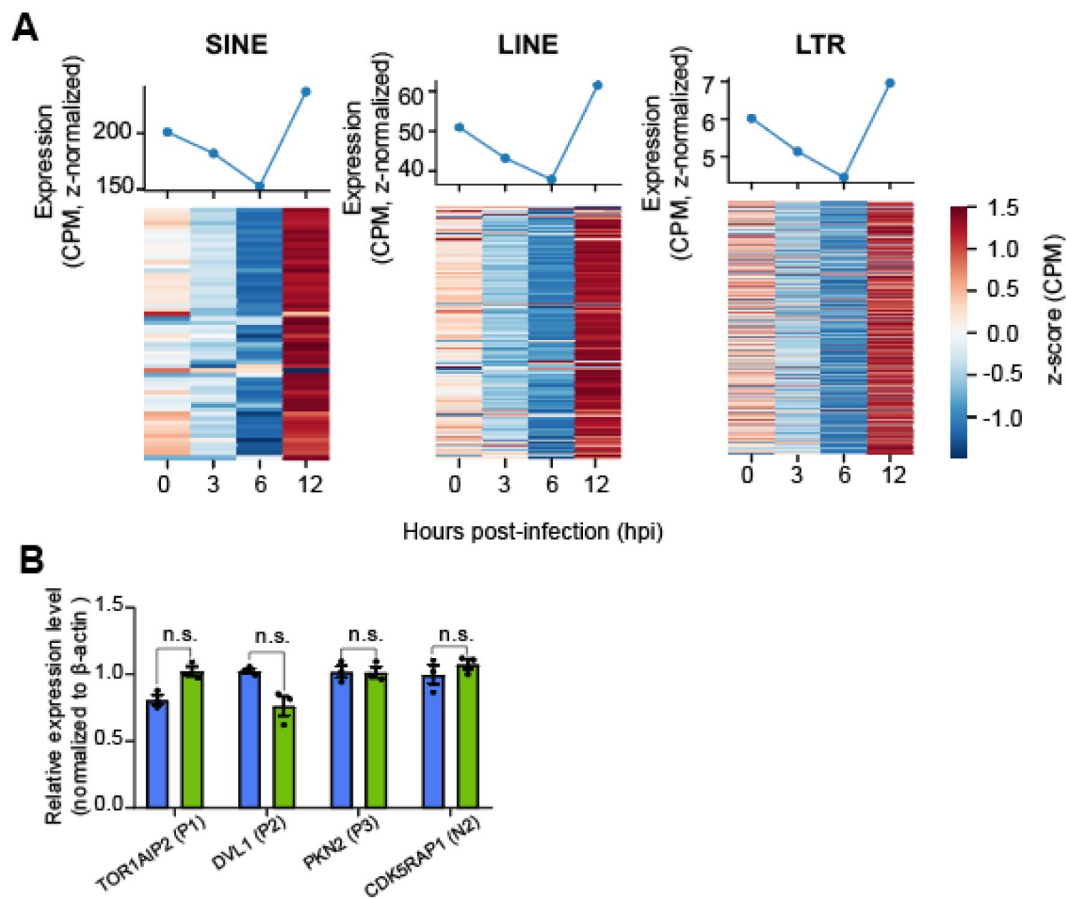
(A-C), Genome browser screenshot over the P1 (A), P2 (B), and P3 (C) HIV-1 induced R-loop-positive chromosomal regions showing result from DRIPc-seq in HIV-1-infected HeLa cells (blue, 0 hpi; yellow, 3 hpi; green, 6 hpi; red, 12 hpi; black, input signals for each indicated sample) on plus (+) or minus (-) DNA strand. Magenta dotted lines represent primer binding sites in qPCR following DRIP. (D and E), Genome browser screenshot over the N1 (D), and N2 (E) HIV-1 induced R-loop-negative chromosomal regions showing result from DRIPc-seq in HIV-1-infected HeLa cells (blue, 0 hpi; yellow, 3 hpi; green, 6 hpi; red, 12 hpi; black, input signals for each indicated sample) on plus (+) or minus (-) DNA strand. Magenta dotted lines represent primer binding sites in qPCR following DRIP.



S3 Fig.

Host cellular R-loop induction by HIV-1 infection is host-genome specific.

(A) DRIP-qPCR using the anti-S9.6 antibody at P1, P2, P3, N1, and N2 in HIV-1-infected cells with MOI of 0.6 harvested at the indicated hpi (blue, 0 hpi; green, 6 hpi). Pre-immunoprecipitated materials were untreated (–) or treated (+) with RNase H, as indicated. Data are presented as the mean $\pm$ SEM; P-values were calculated using one-way ANOVA ($n = 2$). (B) Dot blot analysis of the R-loop in gDNA extracts from HIV-1 infected HeLa cells with MOI of 0.6 harvested at 6hpi. The cells were treated with DMSO, 10uM of Nevirapine (NVP), or 10uM of Raltegravir (RAL) for 24 h before infection, as indicated. gDNAs were probed with anti-S9.6. gDNA extracts were incubated with or without RNase H in vitro before membrane loading (anti-RNA/DNA signal). Fold-induction was normalized to the value of harvested cells at 0 hpi by quantifying the dot intensity of the blots and calculating the ratios of the S9.6 signal to the total amount of gDNA (anti-ssDNA signal). (C) Representative images of the immunofluorescence assay of S9.6 nuclear signals in HIV-1 infected HeLa cells with MOI of 0.6 at 6 hpi. The cells were pre-extracted of cytoplasm and co-stained with anti-S9.6 (red), anti-nucleolin antibodies (green), and DAPI (blue). The cells were treated with DMSO, 10uM of Nevirapine (NVP), or 10uM of Raltegravir (RAL) for 24 h before infection, as indicated. Quantification of S9.6 signal intensity per nucleus after nucleolar signal subtraction for the immunofluorescence assay. The mean value for each data point is indicated by the red line. Statistical significance was assessed using one-way ANOVA ($n > 51$). (D) Pie graphs indicating the percentage of DRIP-seq reads aligned to host cellular genome (aquamarine) or to HIV-1 viral genome (gray), out of the total consensus DRIP-seq peaks from HIV-infected HeLa cells, primary CD4⁺ T cells and Jurkat cells.



S4 Fig.

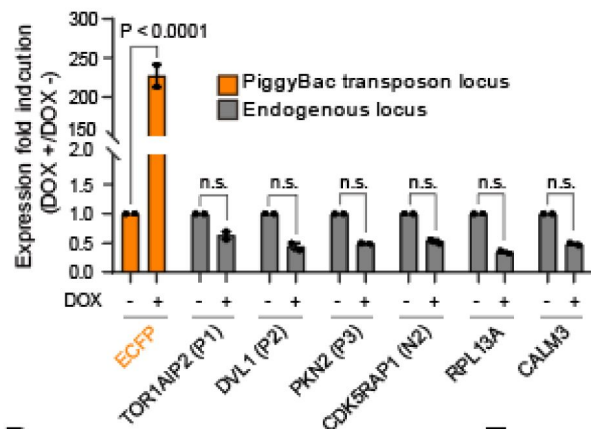
R-loop induction by HIV-1 infection does not follow transcriptome changes in HeLa cells.

(A) Line graphs and heat maps representing expression levels of indicated repetitive elements (SINE, right; LINE, middle; LTR, left) at the indicated hpi of HIV-1 in HeLa cells. Data are presented as the mean expression levels of two biologically independent experiments. (B) Indicated gene expression as measured by RT-qPCR in 0 or 6 hpi harvested HIV-1-infected HeLa cells. Data represent mean $\pm$ SEM, $n = 3$, P values were calculated according to two-tailed Student's t-test. $P > 0.05$; n.s., not significant.

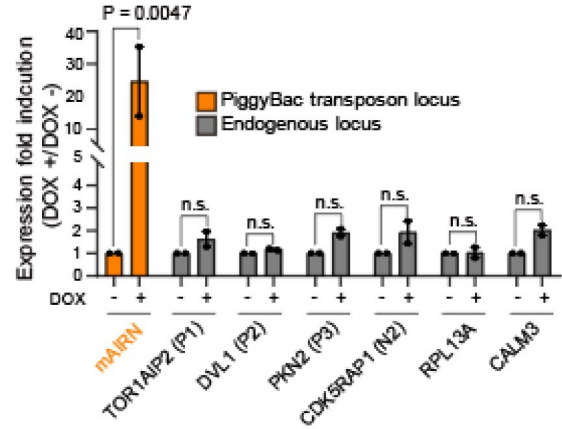
A

Cell line	Pbcopy avg Ct	UCR1 avg Ct	$\Delta\Delta Ct$ $2^{\Delta\Delta Ct}(\text{Pbcopy-UCR1})$	Copy #per Genome ($\Delta\Delta Ct/2$)
pgR-poor	20.15	25.91	54.19	27
pgR-rich	20.12	25.63	45.57	23

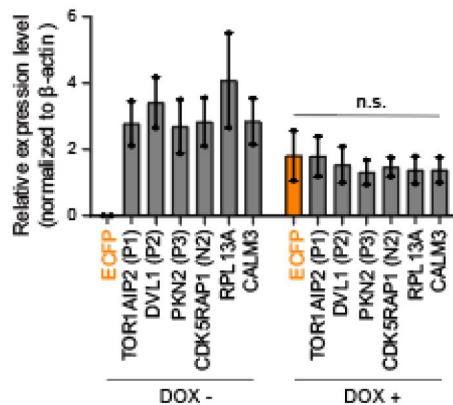
B



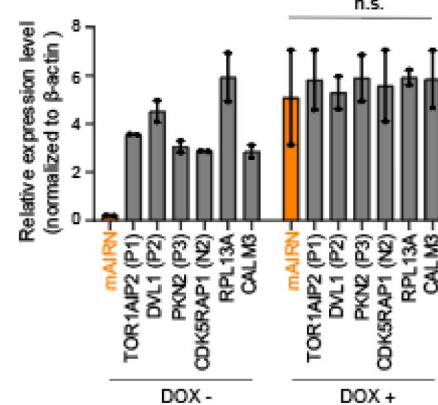
C



D



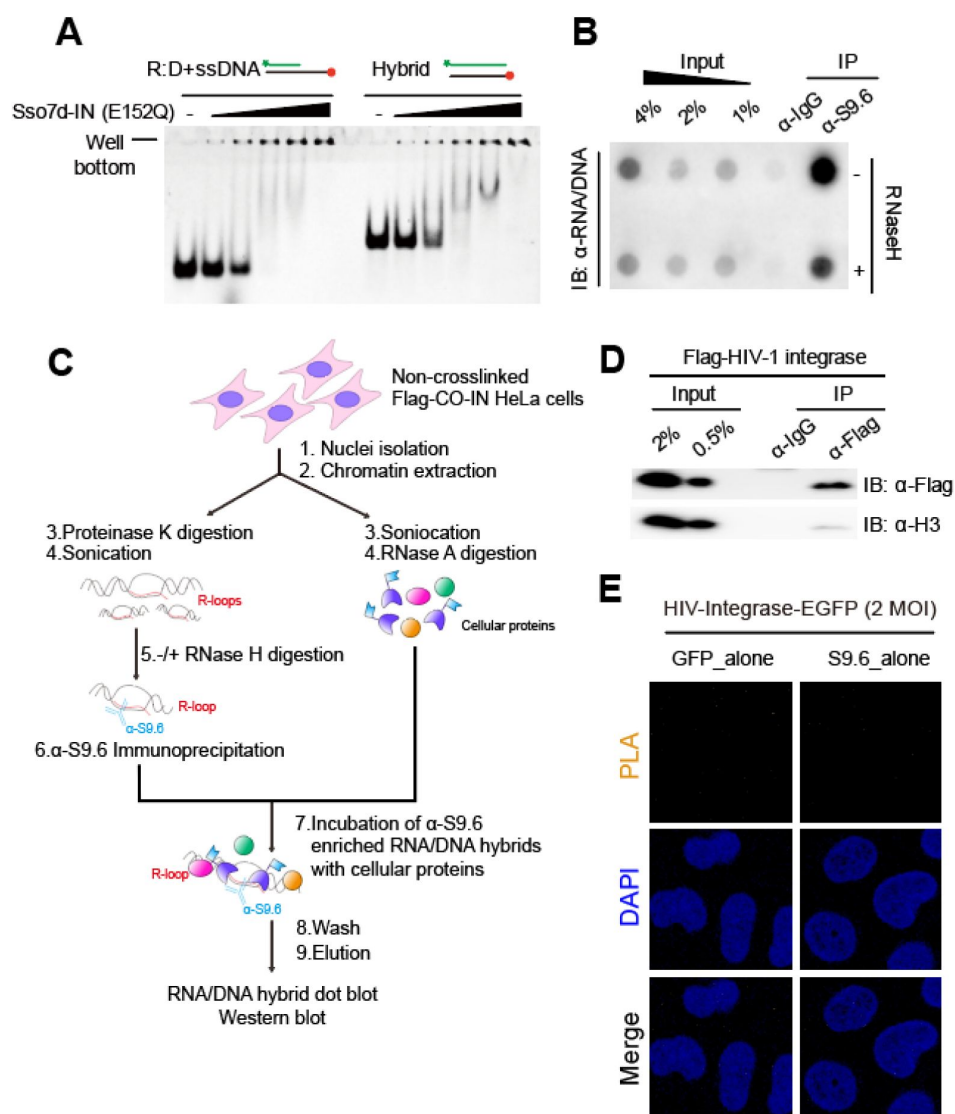
E



S5 Fig.

Regulation of cellular R-loops by RNase H1 expression, or by transposon-transposase insertion of R-loop forming and non-R-loop forming sequences in HeLa cells.

(A) Copy number of piggyBac transposon inserts in each cell line constructed by transfecting the transposon vector and transposase-expressing vector. (B and C) Fold induction of gene expression for the indicated genes as measured by RT-qPCR. Fold induction were calculated by dividing the gene expression level of DOX-treated (+) by that of DOX-untreated (-) in pgR-poor cells (B) or pgR-rich cells (C). Data represent mean $\pm$ SEM, $n = 2$, P values were calculated according to two-way ANOVA. $P > 0.05$; n.s., not significant. (D and E) Relative gene expression of the indicated genes as measured by RT-qPCR in DOX-treated (+) or DOX-untreated (-) pgR-poor cells (D) or pgR-rich cells (E). Data represent mean $\pm$ SEM, $n = 2$, P values were calculated according to two-way ANOVA. $P > 0.05$; n.s., not significant.



S6 Fig.

HIV-1 integrase proteins directly binds to host genomic R-loops.

(A) Representative gel images for EMSA of Sso7d-tagged HIV-1-integrase (E152Q) with different types of nucleic acid substrates (R:D+ssDNA and Hybrid). 100 nM nucleic acid substrate was incubated with Sso7d-tagged HIV-1-integrase (E152Q) at 0 nM, 20 nM, 50 nM, 100 nM, 200 nM, and 400 nM ($n = 3$). (B) Nucleic acid extracts from FLAG-HIV-1-integrase-expressing cells were immunoprecipitated using S9.6 antibody. gDNA was precipitated from the elutes of immunoprecipitation and subjected to DNA-RNA hybrid dot blotting. Where indicated, the gDNA extracts were either untreated (-) or treated (+) with RNase H after elution of immunoprecipitated materials. (C) Summary of the experimental design for R-loop immunoprecipitation using S9.6 antibody in FLAG-tagged HIV-1 integrase protein-expressing HeLa cells with pre-immunoprecipitation in vitro RNase H treatment. (D) Protein extracts from FLAG-HIV-1-integrase-expressing cells were immunoprecipitated using anti-FLAG antibody. Western blot of FLAG immunoprecipitation was probed with anti-FLAG or anti-H3 antibodies. (E) Representative images of the proximity-ligation assay (PLA) using single antibody (anti-GFP or anti-S9.6) in HIV-IN-EGFP virion-infected HeLa cells at 6 hpi, as PLA signal negative controls. Cells were subjected to PLA (orange) and co-stained with DAPI (blue) ($n > 50$).

HeLa				
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
RPL13A	chr19	49487608-49493057	Input (-)_0hpi	3.59
			Input (-)_3hpi	0.24
			Input (-)_6hpi	2.39
			Input (-)_12hpi	3.51
			Input (+)_0hpi	82.29
			Input (+)_3hpi	51.76
			Input (+)_6hpi	39.14
			Input (+)_12hpi	176.73
			IP_RNase H- (-)_0hpi	2.21
			IP_RNase H- (-)_3hpi	2.73
			IP_RNase H- (-)_6hpi	2.39
			IP_RNase H- (-)_12hpi	4.25
			IP_RNase H- (+)_0hpi	110.32
			IP_RNase H- (+)_3hpi	140.22
			IP_RNase H- (+)_6hpi	58.36
			IP_RNase H- (+)_12hpi	137.37
			IP_RNase H+ (-)_0hpi	0.00
			IP_RNase H+ (-)_3hpi	4.48
			IP_RNase H+ (-)_6hpi	3.74
			IP_RNase H+ (-)_12hpi	0.00
			IP_RNase H+ (+)_0hpi	1.98
			IP_RNase H+ (+)_3hpi	3.36
			IP_RNase H+ (+)_6hpi	1.60
			IP_RNase H+ (+)_12hpi	6.81
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
CALM3	chr19	46601330-46610782	Input (-)_0hpi	1.47
			Input (-)_3hpi	1.02
			Input (-)_6hpi	2.46
			Input (-)_12hpi	0.74
			Input (+)_0hpi	26.50
			Input (+)_3hpi	19.95
			Input (+)_6hpi	11.61
			Input (+)_12hpi	56.92
			IP_RNase H- (-)_0hpi	0.90
			IP_RNase H- (-)_3hpi	1.54
			IP_RNase H- (-)_6hpi	1.23
			IP_RNase H- (-)_12hpi	1.73
			IP_RNase H- (+)_0hpi	13.97
			IP_RNase H- (+)_3hpi	28.68
			IP_RNase H- (+)_6hpi	10.58
			IP_RNase H- (+)_12hpi	24.70
			IP_RNase H+ (-)_0hpi	0.71
			IP_RNase H+ (-)_3hpi	1.83
			IP_RNase H+ (-)_6hpi	2.78
			IP_RNase H+ (-)_12hpi	1.04
			IP_RNase H+ (+)_0hpi	2.12
			IP_RNase H+ (+)_3hpi	1.64
			IP_RNase H+ (+)_6hpi	2.26
			IP_RNase H+ (+)_12hpi	1.65
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
SNRPN	chr15	24823647-24978582	Input (-)_0hpi	1.46
			Input (-)_3hpi	1.27
			Input (-)_6hpi	1.34
			Input (-)_12hpi	1.76
			Input (+)_0hpi	1.21
			Input (+)_3hpi	0.81
			Input (+)_6hpi	1.25
			Input (+)_12hpi	0.41
			IP_RNase H- (-)_0hpi	0.45
			IP_RNase H- (-)_3hpi	0.47
			IP_RNase H- (-)_6hpi	0.37
			IP_RNase H- (-)_12hpi	0.05
			IP_RNase H- (+)_0hpi	0.37
			IP_RNase H- (+)_3hpi	0.24
			IP_RNase H- (+)_6hpi	0.54
			IP_RNase H- (+)_12hpi	0.07
			IP_RNase H+ (-)_0hpi	1.40
			IP_RNase H+ (-)_3hpi	0.93
			IP_RNase H+ (-)_6hpi	1.10
			IP_RNase H+ (-)_12hpi	1.31
			IP_RNase H+ (+)_0hpi	1.18
			IP_RNase H+ (+)_3hpi	1.12
			IP_RNase H+ (+)_6hpi	1.26
			IP_RNase H+ (+)_12hpi	1.10

S1 Table.

Chromosomal position and DRIPc-seq signal for referenced R-loop-positive and -negative regions in HIV-1 infected HeLa cells

CD4+				
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
RPL13A	chr19	49487608-49493057	Input (-)_0hpi	2.33
			Input (-)_3hpi	1.51
			Input (-)_6hpi	2.56
			Input (-)_12hpi	0.77
			Input (+)_0hpi	2.91
			Input (+)_3hpi	1.94
			Input (+)_6hpi	2.36
			Input (+)_12hpi	2.19
			IP_RNase H- (-)_0hpi	0.00
			IP_RNase H- (-)_3hpi	3.63
			IP_RNase H- (-)_6hpi	0.00
			IP_RNase H- (-)_12hpi	0.00
			IP_RNase H- (+)_0hpi	144.19
			IP_RNase H- (+)_3hpi	77.26
			IP_RNase H- (+)_6hpi	130.86
			IP_RNase H- (+)_12hpi	190.08
			IP_RNase H+ (-)_0hpi	1.42
			IP_RNase H+ (-)_3hpi	0.00
			IP_RNase H+ (-)_6hpi	0.00
			IP_RNase H+ (-)_12hpi	0.00
			IP_RNase H+ (+)_0hpi	0.93
			IP_RNase H+ (+)_3hpi	0.00
			IP_RNase H+ (+)_6hpi	0.00
			IP_RNase H+ (+)_12hpi	2.28
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
CALM3	chr19	46601330-46610782	Input (-)_0hpi	4.58
			Input (-)_3hpi	4.64
			Input (-)_6hpi	2.96
			Input (-)_12hpi	4.04
			Input (+)_0hpi	3.62
			Input (+)_3hpi	3.65
			Input (+)_6hpi	3.40
			Input (+)_12hpi	4.11
			IP_RNase H- (-)_0hpi	0.00
			IP_RNase H- (-)_3hpi	0.00
			IP_RNase H- (-)_6hpi	0.00
			IP_RNase H- (-)_12hpi	2.70
			IP_RNase H- (+)_0hpi	108.23
			IP_RNase H- (+)_3hpi	183.80
			IP_RNase H- (+)_6hpi	87.73
			IP_RNase H- (+)_12hpi	181.80
			IP_RNase H+ (-)_0hpi	2.80
			IP_RNase H+ (-)_3hpi	0.00
			IP_RNase H+ (-)_6hpi	0.00
			IP_RNase H+ (-)_12hpi	1.94
			IP_RNase H+ (+)_0hpi	4.11
			IP_RNase H+ (+)_3hpi	1.19
			IP_RNase H+ (+)_6hpi	9.88
			IP_RNase H+ (+)_12hpi	6.17
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
SNRPN	chr15	24823647-24978582	Input (-)_0hpi	1.65
			Input (-)_3hpi	1.41
			Input (-)_6hpi	1.74
			Input (-)_12hpi	1.15
			Input (+)_0hpi	1.72
			Input (+)_3hpi	1.46
			Input (+)_6hpi	1.97
			Input (+)_12hpi	1.29
			IP_RNase H- (-)_0hpi	0.31
			IP_RNase H- (-)_3hpi	0.27
			IP_RNase H- (-)_6hpi	0.10
			IP_RNase H- (-)_12hpi	0.27
			IP_RNase H- (+)_0hpi	0.98
			IP_RNase H- (+)_3hpi	1.00
			IP_RNase H- (+)_6hpi	0.53
			IP_RNase H- (+)_12hpi	0.56
			IP_RNase H+ (-)_0hpi	0.94
			IP_RNase H+ (-)_3hpi	1.57
			IP_RNase H+ (-)_6hpi	0.00
			IP_RNase H+ (-)_12hpi	2.17
			IP_RNase H+ (+)_0hpi	1.37
			IP_RNase H+ (+)_3hpi	1.14
			IP_RNase H+ (+)_6hpi	1.42
			IP_RNase H+ (+)_12hpi	1.19

S2 Table.

Chromosomal position and DRIPc-seq signal for referenced R-loop-positive and -negative regions in HIV-1 infected primary CD4⁺ T cells

Jurkat				
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
RPL13A	chr19	49487608-49493057	Input (-)_0hpi	1.46
			Input (-)_3hpi	1.92
			Input (-)_6hpi	1.92
			Input (-)_12hpi	1.58
			Input (+)_0hpi	1.40
			Input (+)_3hpi	2.02
			Input (+)_6hpi	1.15
			Input (+)_12hpi	1.54
			IP_RNase H- (-)_0hpi	0.00
			IP_RNase H- (-)_3hpi	10.17
			IP_RNase H- (-)_6hpi	9.60
			IP_RNase H- (-)_12hpi	2.64
			IP_RNase H- (+)_0hpi	404.40
			IP_RNase H- (+)_3hpi	183.88
			IP_RNase H- (+)_6hpi	486.50
			IP_RNase H- (+)_12hpi	526.25
			IP_RNase H+ (-)_0hpi	0.00
			IP_RNase H+ (-)_3hpi	3.53
			IP_RNase H+ (-)_6hpi	0.00
			IP_RNase H+ (-)_12hpi	0.00
			IP_RNase H+ (+)_0hpi	6.13
			IP_RNase H+ (+)_3hpi	0.00
			IP_RNase H+ (+)_6hpi	0.00
			IP_RNase H+ (+)_12hpi	0.00
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
CALM3	chr19	46601330-46610782	Input (-)_0hpi	2.40
			Input (-)_3hpi	2.18
			Input (-)_6hpi	2.26
			Input (-)_12hpi	2.78
			Input (+)_0hpi	2.08
			Input (+)_3hpi	2.78
			Input (+)_6hpi	1.99
			Input (+)_12hpi	2.38
			IP_RNase H- (-)_0hpi	0.00
			IP_RNase H- (-)_3hpi	11.73
			IP_RNase H- (-)_6hpi	5.58
			IP_RNase H- (-)_12hpi	5.22
			IP_RNase H- (+)_0hpi	208.25
			IP_RNase H- (+)_3hpi	182.67
			IP_RNase H- (+)_6hpi	167.98
			IP_RNase H- (+)_12hpi	220.30
			IP_RNase H+ (-)_0hpi	0.00
			IP_RNase H+ (-)_3hpi	2.04
			IP_RNase H+ (-)_6hpi	0.00
			IP_RNase H+ (-)_12hpi	4.84
			IP_RNase H+ (+)_0hpi	13.84
			IP_RNase H+ (+)_3hpi	1.62
			IP_RNase H+ (+)_6hpi	4.37
			IP_RNase H+ (+)_12hpi	3.29
Gene	Chromosom	Position (hg38)	Description	Average DRIPc-seq signal
SNRPN	chr15	24823647-24978582	Input (-)_0hpi	1.75
			Input (-)_3hpi	1.94
			Input (-)_6hpi	1.87
			Input (-)_12hpi	1.84
			Input (+)_0hpi	1.86
			Input (+)_3hpi	1.89
			Input (+)_6hpi	1.81
			Input (+)_12hpi	1.73
			IP_RNase H- (-)_0hpi	0.12
			IP_RNase H- (-)_3hpi	0.00
			IP_RNase H- (-)_6hpi	0.17
			IP_RNase H- (-)_12hpi	0.00
			IP_RNase H- (+)_0hpi	2.43
			IP_RNase H- (+)_3hpi	2.19
			IP_RNase H- (+)_6hpi	2.23
			IP_RNase H- (+)_12hpi	2.36
			IP_RNase H+ (-)_0hpi	2.58
			IP_RNase H+ (-)_3hpi	3.46
			IP_RNase H+ (-)_6hpi	1.62
			IP_RNase H+ (-)_12hpi	1.87
			IP_RNase H+ (+)_0hpi	1.78
			IP_RNase H+ (+)_3hpi	2.38
			IP_RNase H+ (+)_6hpi	1.06
			IP_RNase H+ (+)_12hpi	1.43

S3 Table.

Chromosomal position and DRIPc-seq signal for referenced R-loop-positive and -negative regions in HIV-1 infected Jurkat cells

	gene_symbol	0hpi	3hpi	6hpi	12hpi	48hpi
P1	TOR1AIP2	1	1.163832	1.247899	1.024926	0.619497
P2	DVL1	1	0.781593	0.571348	0.901502	0.270459
P3	PKN2	1	1.280974	1.891552	1.31842	1.515107
N1	N/A	N/A	N/A	N/A	N/A	N/A
N2	CDK5RAP1	1	0.73977	0.775	1.143662	0.472377

S4 Table.

RNA-seq analysis of relative gene expression levels of P1-3 and N1,2 R-loop regions

Oligonucleotides	Sequence 5' to 3'	Remark
PCR primer 1.0 P5	AATGATACGGCGACCCAGCATCTACACTCTTTCCCTACACGA	amplification primer
PCR primer 2.0 P7	CAAGAGAAGACGGCATAAGAGAT	amplification primer
Index Adapter 1	GATCGGAAGAGCACACGTCTGAACTCCAGTCACATCACGATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi Input replicate 1
Index Adapter 2	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGATGATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi Input replicate 1
Index Adapter 3	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCTAGGCAATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi Input replicate 1
Index Adapter 4	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGATCCTGATGCCGTCTTCTGCTTG	HeLa 12hpi Input replicate 1
Index Adapter 5	GATCGGAAGAGCACACGTCTGAACTCCAGTCACAGTGAATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi RNH-IP replicate 1
Index Adapter 6	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGCCAATATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi RNH-IP replicate 1
Index Adapter 7	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGATCATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi RNH-IP replicate 1
Index Adapter 8	GATCGGAAGAGCACACGTCTGAACTCCAGTCACAGTCAACATTCGATGCCGTCTTCTGCTTG	HeLa 12hpi RNH-IP replicate 1
Index Adapter 9	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGATCAGATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi RNH+IP replicate 1
Index Adapter 10	GATCGGAAGAGCACACGTCTGAACTCCAGTCACAGTCACTAGCTTATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi RNH+IP replicate 1
Index Adapter 11	GATCGGAAGAGCACACGTCTGAACTCCAGTCACGGGTACATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi RNH+IP replicate 1
Index Adapter 12	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCTTGTAACTCTCGTATGCCGTCTTCTGCTTG	HeLa 12hpi RNH+IP replicate 1
Index Adapter 28	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCAAAATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi Input replicate 2
Index Adapter 29	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCAACTAATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi Input replicate 2
Index Adapter 30	GATCGGAAGAGCACACGTCTGAACTCCAGTCACACCGGATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi Input replicate 2
Index Adapter 31	GATCGGAAGAGCACACGTCTGAACTCCAGTCACACGATATCTCGTATGCCGTCTTCTGCTTG	HeLa 12hpi Input replicate 2
Index Adapter 32	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCACTCAATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi RNH-IP replicate 2
Index Adapter 33	GATCGGAAGAGCACACGTCTGAACTCCAGTCACAGCGGATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi RNH-IP replicate 2
Index Adapter 34	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCATTTATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi RNH-IP replicate 2
Index Adapter 35	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCAAAATCTCGTATGCCGTCTTCTGCTTG	HeLa 12hpi RNH-IP replicate 2
Index Adapter 36	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCGAATATCTCGTATGCCGTCTTCTGCTTG	HeLa 0hpi RNH+IP replicate 2
Index Adapter 37	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCGAATATCTCGTATGCCGTCTTCTGCTTG	HeLa 3hpi RNH+IP replicate 2
Index Adapter 38	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCTAGCTATCTCGTATGCCGTCTTCTGCTTG	HeLa 6hpi RNH+IP replicate 2
Index Adapter 39	GATCGGAAGAGCACACGTCTGAACTCCAGTCACCTATACATCTCGTATGCCGTCTTCTGCTTG	HeLa 12hpi RNH+IP replicate 2

S5 Table.

Oligonucleotides used for DRIPc-seq library construction

Oligonucleotides	Sequence 5' to 3'
P1 Fwd	TTATAAGTCAGCCTCCAGGATCAA
P1 Rev	TTCAGGTCTAGGCAGTCTGA
P2 Fwd	GGA CAG ATG ACA GGG TCG C
P2 Rev	ATG AGG AAG ACC CCC TCG G
P3 Fwd	CTCTGTGTAACGCTGGTGCT
P3 Rev	ACACGCTTCTGACCACTAAGG
N1 Fwd	TTG GCC CTA CTG AAT GAT TGG T
N1 Rev	TTA AGG CAT GCT CAG GCG A
N2 Fwd	TGA GAT TTC AGG TTC CAT GAT TTG
N2 Rev	TGC TCA GTG TTC TAA TTT CCC TGT
β -actin Fwd	AGAGCTACGAGCTGCCTGAC
β -actin Rev	AGCACTGTGTTGGCGTACAG
SH49 (ECFP Fwd)	TGGTTTGTCCAACTCATCAA
SH40 (mAIRN Fwd)	CGAGAGAGGCTAAGGGTGAA
SH21 (ECFP/mAIRN Rev)	ACATGGTCCTGCTGGAGTTC
RT-qPCR P1 (TOR1AIP2) Fwd	CCTTGGTCTTTCCCACTTGAGTG
RT-qPCR P1 (TOR1AIP2) Rev	GCAGGGTTAAAACCAGCTACTCG
RT-qPCR P2 (DVL1) Fwd	GCATAACCGACTCCACCATGTC
RT-qPCR P2 (DVL1) Rev	GATGGAGCCAATGTAGATGCCG
RT-qPCR P3 (PKN2) Fwd	GCATCACCAACACTAAGTCCACG
RT-qPCR P3 (PKN2) Rev	GCTTTTGACCGTCCAGGGACAT
RT-qPCR N2 (CDK5RAP1) Fwd	AGAGTGGAAGCAGCCGTGTGTT
RT-qPCR N2 (CDK5RAP1) Rev	GATCTTCCTCCGTCTCACCACA

S6 Table.

Primers used for qPCR

Oligonucleotides	Sequence 5' to 3'	Remark
R-loop oligo1*	5'-[Cy3]-GCC AGG GAC GAG GTG AAC CTG CAG GTG GGC GGC TAC TAC TTA GAT GTC ATC CGA GGC TTA TTG GTA GAA TTC GGC AGC GTC ATG C GA CGG C-3'	R-loop, R:D+ ssDNA, ssDNA
R-loop oligo2*	5'-GCC GTC GCA TGA CGC TGC CGA ATT CTA CCA CGC GAT TCA TAC CTG TCG TGC CAG CTG CTT TGC CCA CCT GCA GGT TCA CCT CGT CCC TGG C-3'	R-loop, dsDNA
R-loop RNA	5'-[Cy5]-GCA GCU GGC ACG ACA GGU AUG AAU C-3'	R-loop, R:D+ssDNA, ssRNA
Homoduplex	5'-[Cy3]-GCC AGG GAC GAG GTG AAC CTG CAG GTG GGC AAA GCA GCT GGC ACG ACA GGT ATG AAT CGC GTG GTA GAA TTC GGC AGC GTC ATG CGA CGG C-3'	dsDNA
Hybrid DNA	5'-CCC ATA CCG TAT AAC CAT TTG GCT GTC CAA GCT CCG GGT-3'	Hybrid
Hybrid RNA	5'-[Cy5]-ACC CGG AGC UUG GAC AGC CAA AUG GUU AUA CGG UAU GGG-3'	Hybrid
oligo 5	5'GCAGTAGCATGACGCTGCTGAATTCACCACGCTATGCT CTCGTCTAGGTTCACTCCGTCCCTGCGATTACCTGTCGTGCCAGCTGC	R:D+ssDNA

S8 Table.

Oligonucleotides used for electrophoretic mobility shift assay substrate preparation

Data	Accession Number/ Website
Jurkat integration site	SRR12322252
TSA-seq_SPAD	SRR3538917, SRR3538918, SRR3538919, SRR3538920
SPIN (Spatial Position Inference of the Nuclear genome) annotation of speckle	https://github.com/ma-compbio/SPIN
LADs	GSE22428

S9 Table.

Accession numbers and data sources.

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Reviewer #1 (Public review):

(1) Significance of findings and strength of evidence.

(a) The work presented in this manuscript is intended to support the authors' novel idea that HIV DNA integration strongly favors "triple-stranded" R-loops in DNA formed either during transcription of many, but not all, genes or by strand invasion of silent DNA by transcripts made elsewhere, and that HIV infection promotes R-loop formation mediated by incoming virions in the absence of reverse transcription. The authors were able to demonstrate a reverse transcription-independent increase in R-loop formation early during HIV infection, while also demonstrating increased integration into sequences that contain R-loop structures. Furthermore, this manuscript also identifies that R-loops are present in both transcriptional active and silent regions of the genome and that HIV integrase interacts with R-loops. Although the work presented supports a correlation between R-loop formation and HIV DNA

integration, it does not prove the authors' hypothesis that R-loops are directly targeted for integration. Direct experimentation, such as in vitro integration into defined DNA targets, will be required. Further, the authors provide no explanation as to how current sophisticated structural models of concerted retroviral DNA integration into both strands of double-stranded DNA targets can accommodate triple-stranded structures. Finally, there are serious technical concerns with interpretation of the integration site analyses.

This resubmitted manuscript has corrected some of the issues raised by the previous reviews - particularly the quality of the English - but otherwise the text and figures remain very much the same and concerns regarding the conclusions drawn regarding integration site specificity remain. The manuscript also still suffers from a lack of description of experimental detail necessary to understand the results as presented. In many cases, explanations given privately in the rebuttal of the earlier reviews need to be made available to all readers, not just the reviewers.

(2) Public review with guidance for readers around how to interpret the work, highlighting important findings but also mentioning caveats.

(a) Introduction: The authors provide an excellent introduction to R-loops but they base the rationale for this study on mis-citation of earlier studies regarding integration in transcriptionally silent regions of the genome. The "most favored locus" cited in the very old reference 6 comprises only 5 events and has not been reproduced in more recent, much larger datasets. For example, see the study of over 300,000 sites in ref 14. The laundry list of IN interactors in lines 43-44 is based on old experiments. It is now quite clear that the only direct interaction of importance is with LEDGF and that should be discussed here. Also discussed should be the role of the capsid in the nuclear entry and targeting. For example, one of the references cited, as well as a mention in the discussion (Line 326) concerns CPSF-6, which is now known to modulate nuclear entry and specificity by interacting with capsid, not integrase. The statement on lines 46-47 regarding that some highly expressed genes are, nonetheless, poor targets for integration is correct, but the experiment cited was done in PBMC with wild-type HIV-1 and it is possible that those genes were expressed in non-target cells like B-cells or monocytes.

(b) Figure 1: Demonstrates models for HIV infections in both cell lines and primary human CD4⁺ T cells. R-loop formation was determined through a method called DRIP-seq which utilizes an anti-body specific for DNA-RNA hybrid structures and sequences these regions of the genome using RNaseH treatment to show that when RNA-DNA hybrids are absent then no R-loops are detected. In these models of in vitro and ex vivo infection, the authors show that R-Loop formation increases following HIV infection between 6 hr. post-infection and 12 hrs. post-infection, depending on the cell model. However, these figures lack a mock infected control for each cell model to assess R-loop formation at the same time points. They would also benefit from a control showing that virus entry is necessary, such as omitting the VSV G protein donor.

(c) Figure 2: This figure shows that cells infected with HIV show more R-loops as well as longer sequences containing R-loop structures. Panel B shows that these R-loops were distributed throughout different genomic features, such as both genic and intergenic regions of the genome. However, the data are presented in such a way that it is impossible to determine the proportion of R-loops in each type of genomic feature. The reader has no way to tell, for example, the proportion of R-loops in genic vs intergenic DNA and how this value changes with time. Furthermore, increased R-loop formation due to HIV infection showed poor correlation with gene expression, suggesting that R-loops were not forming due to transcriptional activation, although the difference between 0 and the remaining timepoints is not apparent, nor is the meaning of the absurd p values.

The experiments presented in Figures 1 and 2 show that treatment of cells with VSV G-pseudotyped HIV-1 leads to a significant increase in R loops in all parts of the genome.

Accumulation of R-loops at so soon after infection, as well as its resistance to RT and Integration inhibitors, rules out the involvement of newly synthesized viral DNA or any newly made viral protein (Figure S3). Rather, some component(s) of the virion, possibly protease, or an accessory gene product such as Vpr or Vif, must be directly responsible (although the authors neglect to draw this conclusion in the description of these experiments, lines 125-135, leaving it hanging until the Discussion).

On the whole, and as a non-expert in this area, I find the overall conclusions of this part of the study convincing, but, as pointed out in one of the earlier reviews, the virologic significance of early effects seen at high multiplicity of infection (likely hundreds of particles per cell) needs to be taken with a grain of salt. At a minimum, this point should be discussed. Also, the study would have been greatly strengthened by a simple experiment to identify the virion protein responsible for the effect.

Based on the results in the first two figures, the authors hypothesize that R-Loop induction early in infection plays an important role in HIV replication, specifically by interacting with the intasome and thus directing integration to regions of the host genome favorable for expression of the provirus. Experiments to test this idea and probe the mechanism are described in the remaining 3 figures, which, despite comments in the previous reviews, are unchanged from the previous version and still suffer from serious defects in experimental design and interpretation.

(d) Figure 3: This figure shows the use of cell lines carrying R-loop inducible (mAIRN) or non-inducible (ECFP) genes to model association of HIV integration with R-loop structures. The authors demonstrate the functional validation of R-loop induction in the cell line model. Additionally, when R-loops are induced there is a significant increase in HIV integration in the R-loop forming vector sequence when R-loops are induced with doxycycline. This result shows a correlation between expression and integration that is much stronger in the R-loop forming gene than in the unreferenced ECFP gene but does not prove that integration directly targets R-loops. It is possible, for example, that some feature of the DNA sequence, such as base composition affects both integration and R-loop formation independently. As described more fully below, there is also a serious concern regarding the method used to quantitate the integration frequencies. As before, There are a number of problems here.

(1) The authors use a classic, but suboptimal integration site assay comprising restriction enzyme digestion followed by PCR to assess integration site distribution, and (despite statements to the contrary in the rebuttal) read counts to quantitate relative frequencies of target site use. See the legend and axis labels in Fig 3E, F, and G. This approach leads to serious bias in the ability to detect and count the use of integration sites that are either too close or too far from the sites of cleavage and can lead to artefactual misrepresentation of their chromosomal distribution.

(2) The result shown in Figure 3D is uninterpretable. It is simply not possible that the 3-fold increase in luciferase activity is due addition of 25 10-kb sequences leading to A 3-fold increase in integration frequency into the target sequence, particularly when panel E shows that the measured frequency is on the order of 20 reads per million. Something else must be going on here.

(3) Panels 3F and G show the read count distribution in the introduced target sequences plotted in a completely nonstandard way and is explained so poorly that I could not be sure what the authors were trying to show. The numbers on the bottom of the 2 plots appear to represent the only sites of integration seen in the 10-kb region studied. If so, this is not the expected result for the authors claim of greatly increasing regional integration. As can easily be seen in the figures of ref 14, high frequency gene targets are characterized by large numbers of sites, not by more frequent targeting of small numbers of sites as implied by the figures.

(e) Figure 4: This figure shows evidence of increased HIV integration within regions of the genome containing R-loops with additional preference with integration within the R-loop and

decrease in frequency of integration further from the R-loop. Identifying a preference for R-loops is very intriguing but the authors do also demonstrate that integration does occur when R-loops are not present. Also Panel A, which shows that regions of cell DNA that form R-loops have a higher frequency of Integration sites than those that do not, should also be controlled for the level of gene expression of the two types of region. the result shown cannot be interpreted to mean that R-loops have anything to do with integration targeting. It is already well-established that about 80% of HIV integration sites are in expressed genes, which comprise about 20% of the genome. Since a gene must be expressed to contain an R-loop, the non-R-loop fraction will contain the 80% of the genome that is a 20-fold poorer target, giving the result shown, whether R-loops are involved or not. The rather weak correlation between R-Loop locations and integration site distribution in Fig 4C and D hardly seems consistent with the curves seen in 4B. Can the authors refute the hypothesis that the apparent correlation is simply because both integration and R-Loop formation frequency must correlate with level of gene expression and therefore their correlation with one another cannot be used to infer causality/ As pointed out in prior reviews, R-loops themselves cannot be targets for integration. In their rebuttal, the authors agree and have made slight modifications to their conclusion in the text, now concluding that Integration favors the vicinity of an R-loop. Why then do the peaks in correlation curves in Fig 4B center exactly on the center of the R-loops? It seems that this result would be more consistent with integration and R-loop formation favoring the same sites, but for different reasons (base composition for example).

(f) Figure 5: In this figure the authors demonstrate that HIV integrase binds to R-loops through a number of protein assays, but does not show that this binding is associated with enzymatic activity. EMSA of integrase identified increased binding to DNA-RNA over dsDNA. Additionally, precipitation of RNA-DNA hybrids pulled down HIV integrase. A proximity ligation assay detecting R-loops and HIV-integrase showed co-localization within the nucleus of HeLa cells. HeLa cells were probably used due to their efficiency of transduction but are not physiologically relevant cell types. Figure 5 suffers greatly in interpretability from the failure of the authors to use assembled intasomes, since the DNA binding properties are likely to be quite different. The authors excuse that they were unable to prepare intasomes (which needs to be included in the text, not just in the rebuttal) explains but does not justify the use of monomeric IN protein. Figure 5A shows that the IN binding is NOT specific to R-loops, since any single-stranded DNA binds equally. The authors should make this point in the text. The experiment using integrase overexpression in cells brings up some déjà vu to a retrovirologist. There is some history in retrovirology of experiments like this having been used to draw conclusions (like the role of integrase in nuclear import) that have since proven to be wrong. Also, Fig 5G is not interpretable quantitatively, since the distribution of neither IN nor R-loops is probed, and we have no idea what proportion of each is in the PLA spots. Overall, this section would be much more convincing if it also included some direct experimentation, such as in vitro integration using intasomes, or infection of cells with viral mutants (or in the presence of inhibitors) affecting the function of whatever virion protein found to be important for R-loop formation.

(g) Discussion: In the discussion, the authors address how their work relates to previous evidence of HIV integration by association of LEDGF/p75 and CPSF6. They also cite that LEDGF/p75 has possible R-loop binding capabilities. They also discuss what possible mechanisms are driving increases in R-loop formation during HIV infection, pointing to possible HIV accessory proteins. They also state that how HIV integrates in transcriptionally silent regions is still unknown but do point out that they were able to show R-loops appear in many different regions of the genome but did not show that R-loops in transcriptional inactive regions are integration targets. More seriously, they failed to make a connection between their work and current understanding of the biochemical and structural mechanism of the integration reaction.

Reviewer #3 (Public review):

In this manuscript, Park and colleagues describe a series of experiments that investigate the role of R-loops in HIV-1 genome integration. The authors show that during HIV-1 infection, R-loops levels on the host genome accumulate. Using a synthetic R-loop prone gene construct, they show that HIV-1 integration sites target sites with high R-loop levels. They further show that integration sites on the endogenous host genome are correlated with sites prone to R-loops. Using biochemical approaches, as well as in vivo co-IP and proximity ligation experiments, the authors show that HIV-1 integrase physically interacts with R-loop structures.

The major strengths of this work is that the investigators use multiple independent experimental systems and multiple cell types to support their conclusions, including in vivo and biochemical experiments. Furthermore, their use of genome-wide analyses help to support their conclusion that HIV targets genomic regions enriched with R-loops versus those lacking such enrichment.

This work may have a significant impact on the field of HIV genomic integration by elucidating why transcription levels are not the sole determinant of HIV integration sites.

<https://doi.org/10.7554/eLife.97348.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

We appreciate the valuable and constructive comments of Reviewer #1 on our manuscript. We have addressed the comments from Reviewer #1 in the public review in the response to the recommendations for the authors, as the public review comments largely overlap with that of the recommendations for the authors.

Reviewer #1 (Recommendations For The Authors):

(1.1) Figure 1 did not use a mock-infected control for the development of R-loops but only a time before infection. I think it would have been a good control to have that after the same time of infection non-infected cells did not show increases in R-loops and this is not a product of the cell cycle.

We prepared our DRIPc-seq library using cell extracts harvested at 0, 3, 6, and 12 h post-infection (hpi), all at the same post-seeding time point. Each sample was infected with HIV-1 virus in a time-dependent manner. Therefore, it is unlikely that the host cellular R-loop induction observed in our DRIPc-seq results was due to R-loop formation during the cell cycle. In Lines 93–95 of the Results section of the revised manuscript, we have provided a more detailed description of our DRIPc-seq library experimental scheme. Thank you.

(1.2) Figure 2 should have included a figure showing the proportion of DRIPc-seq peaks located in different genome features relative to one another instead of whether they were influenced by time post-infection. Figure 2C was performed in HeLa cells, but primary T cell data would have been more relevant as primary CD4+ T cells are more relevant to HIV infection.

We have included a new figure presenting the relative proportion of DRIPc-seq peaks mapped to different genomic features at each hpi (Fig. 2C of the revised manuscript). We found that the proportion of DRIPc-seq peaks mapped to various genomic compartments remained consistent over the hours following the HIV-1 infection. This further supports our original claim that HIV-1 infection does not induce R-loop enrichment at specific genomic features but that the accumulation of R-loops after HIV-1 infection is widely distributed.

We considered HeLa cells as the primary in vitro infection model, therefore, we conducted RNA-seq only on HeLa cells. However, we agree with the reviewer's opinion that data from primary CD4⁺ T cells may be more physiologically relevant. Nevertheless, as demonstrated in the new figure (Fig. 2C of the revised manuscript), HIV-1 infection did not significantly alter the proportion of R-loop peaks mapped to specific genomic compartments, such as gene body regions, in HeLa, primary CD4⁺ T, and Jurkat cells. Therefore, we anticipate no clear correlation between changes in gene expression levels and R-loop peak detection upon HIV-1 infection, even in primary T cells. Thank you.

(1.3) Figure 5G is very hard to see when printed, is there a change in brightness or contrast that could be used? The arrows are helpful but they don't seem to be pointing to much.

We have highlighted the intensity of the PLA foci and magnified the images in Fig. 5G in the revised manuscript. While editing the images according to your suggestion, we found a misannotation regarding the multiplicity of infection in the number of PLA foci per nucleus quantification analysis graph in Fig. 5G of the original manuscript. We have corrected this issue and hope that it is now much clearer.

(1.4) The introduction provided a good background for those who may not have a comprehensive understanding of DNA-RNA hybrids and R-loops, but the rationale that integration in non-expressed sequence implies that R-loops may be involved is very weak and was not addressed experimentally. A better rationale would have been to point out that, although integration in genes is strongly associated with gene expression, the association is not perfect, particularly in that some highly expressed genes are, nonetheless, poor integration targets.

In accordance with the reviewer's comment, we revised the Introduction. We have deleted the statement and reference in the introduction "... the most favored region of HIV-1 integration is an intergenic locus, ...", which may overstate the relevance of the R-loop in HIV-1 integration events in non-expressed sequences. Instead, we introduced a more recent finding that high levels of gene expression do not always predict high levels of integration, together with the corresponding citation (Lines 46–47 of the revised manuscript), according to the reviewer's suggestion in the reviewer's public review 2)-(a).

(1.5) The discussion was seriously lacking in connecting their conclusions regarding R-loop targeting of integration to how integration works at the structural level, where it is very clear that concerted integration on the two DNA strands ca 5 bp apart is essential to correct, 2-ended integration. It is very difficult to visualize how this would be possible with the triple-stranded R-loop as a target. The manuscript would be greatly strengthened by an experiment showing concerted integration into a triplestranded structure in vitro using PICs or pure integrase.

We believe there has been a misunderstanding of our interpretation regarding the putative role of R-loop structures in the HIV-1 integration site mechanism because of some misleading statements in our original manuscript. Based primarily on our current data, we believe that R-loop structures are bound by HIV-1 integrase proteins and lead to HIV-1 viral genome

integration into the vicinity regions of the host genomic R-loops. By carefully revising our manuscript, we found that the title, abstract, and discussion of our original manuscript includes phrases, such as “HIV-1 targets R-loops for integration,” which may overstate our finding on the role of R-loop in HIV-1 integration site selection. We replaced these phrases. For example, we used phrases, such as, “HIV-1 favors vicinity regions of R-loop for the viral genome integration,” in the revised manuscript. We apologize for the inconvenience caused by the unclear and nonspecific details of our findings.

Using multiple biochemical experiments, we successfully demonstrated the interaction between the cellular R-loop and HIV-1 integrase proteins in cells and in vitro (Fig. 5 of the revised manuscript). However, we could not validate whether the center of the triple-stranded R-loops is the extraction site of HIV-1 integration, where the strand transfer reaction by integrase occurs. This is because an R-loop can be multi-kilobase in size (1, 2); therefore, we displayed a large-scale genomic region (30-kb windows) to present the integration sites surrounding the R-loop centers. Nevertheless, we believe that we validated R-loop-mediated HIV-1 integration in R-loop-forming regions using our pgR-poor and pgR-rich cell line models. When infected with HIV-1, pgR-rich cells, but not pgR-poor cells, showed higher infectivity upon R-loop induction in designated regions following DOX treatment (Fig. 3C and 3D of the revised manuscript). In addition, we quantified site-specific integration events in R-loop regions, and found that a greater number of integration events occurred in designated regions of the pgR-rich cellular genome upon R-loop induction by DOX treatment, but not in pgR-poor cells (Fig. 3E–G of the revised manuscript).

We agree with the reviewer that an experiment showing the concerted integration of purified PICs into a triple-stranded structure in vitro would greatly strengthen our manuscript. We attempted the purification of viral DNA (vDNA)-bound PICs using either Sso7d-tagged HIV-1 integrase proteins or non-tagged HIV-1 integrase proteins (F185K/C280S) procured from the NIH HIV reagent program (HRP-20203), following the method described by Passos et al., Science, 2017; 355 (89-92) (3). Despite multiple attempts, we could not purify the nucleic acid-bound protein complexes for in vitro integration assays. However, we believe that pgR-poor and pgR-rich cell line models provide a strong advantage in specificity of our primer readouts. Compounded with our in cellulo observation, we believe that our work provides strong evidence for a causative relationship between R-loop formation/R-loop sites and HIV-1 integration.

Additionally, in the Discussion section of the revised manuscript, we have expanded our discussion on the role of genomic R-loops contributing in molding the host genomic environment for HIV-1 integration site selection, and the potential explanation on how R-loops are driving integration over long-range genomic regions. Thank you.

(1.6) There are serious concerns with the quantitation of integration sites used here, which should be described in detail following line 503 but isn't. In Figure 3, E-G, they are apparently shown as reads per million, while in Figure 4B as "sites (%)" and in 4C as log10 integration frequency." Assuming the authors mean what they say, they are using the worst possible method for quantitation. Counting reads from restriction enzyme-digested, PCR-digested DNA can only mislead. At the numbers provided (MOI 0.6, 10 µg DNA assayed) there would be about 1 million proviruses in the samples assayed, so the probability of any specific site being used more than once is very low, and even less when one considers that a 10% assay efficiency is typical of integration site assays. Although the authors may obtain millions of reads per experiment, the number of reads per site is an irrelevant value, determined only by technical artefacts in the PCR reactions, most significantly the length of the amplicons, a function of the distance from the integration site to the nearest MstII site, further modified by differences in Tm. Better is to collapse identical reads to 1 per site, as may have been done in Figure 4B, however, the efficiency of integration site detection will still be inversely related to the length of the amplicon.

Indeed, if the authors were to plot the read frequency against distance to the nearest MstII site, it is likely that they would get plots much like those in Figure 4B.

Detailed methods for integration site sequencing data processing are described in the Materials and Methods section of the revised manuscript (Line 621–631 of the revised manuscript). We primarily followed HIV-1 integration site sequencing data processing methods previously described by Li et al., mBio, 2020; 11(5) (4).

While it may be correct that the HIV-1 integration event cannot occur more than once at a given site, our Fig. 3E, 4C, and 4D of the revised manuscript present the number of integration-site sequencing read counts expressed in reads-per-million (RPM) units or as log10-normalized values. Based on the number of mapped reads from the integration site sequencing results, we can infer that there was an integration event at this site, whether it was a single or multiple event.

We believe that the original annotation of y-axis, “Integration frequency,” may be misleading as it can be interpreted as a probability of any specific site being used for HIV-1 integration. Therefore, we corrected it as “number of mapped read” for clarity (Fig. 3E–G, 4C and 4D, and the corresponding figure legends of the revised manuscript). We apologize for any confusion. Thank you.

Other points:

(1.7) Overall: There are numerous grammatical and usage errors, especially in agreement of subject and verb, and missing articles, sometimes multiple times in the same sentence. These must be corrected prior to resubmission.

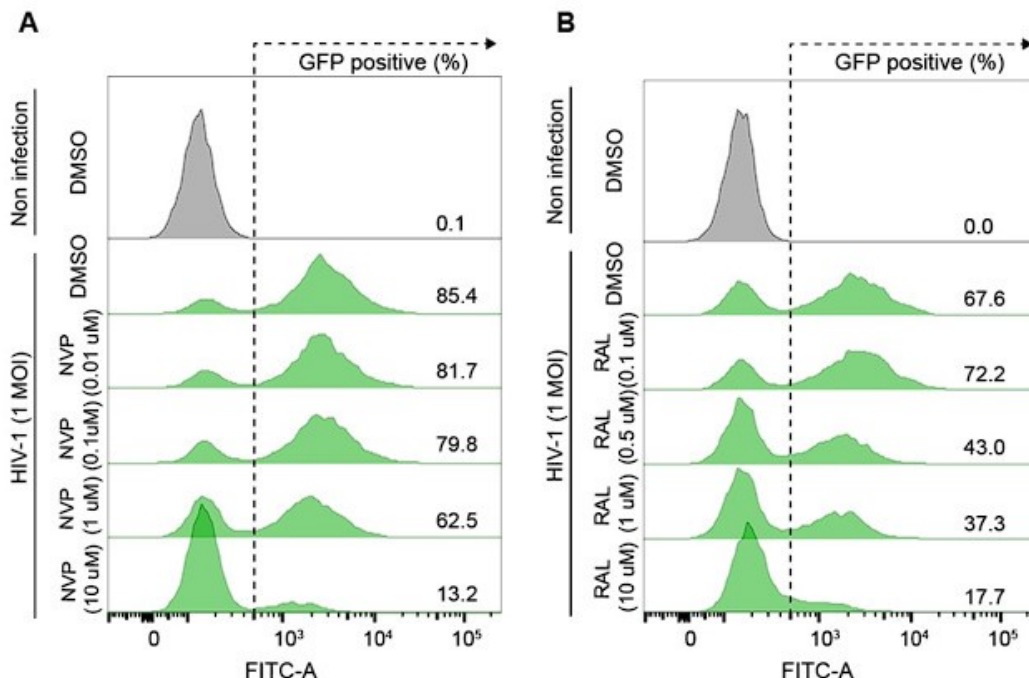
The revised manuscript was edited by a professional editing service. Thank you.

(1.8) Line 126-134: A striking result, but it needs more controls, as discussed above, including a dose-response analysis.

We determined the doses of NVP and RAL inhibitors in HeLa cells by optimizing the minimum dose of drug treatment that provided a sufficient inhibitory effect on HIV1 infection (Author response image 1). The primary objective of this experiment was to determine R-loop formation while reverse transcription or integration of the HIV-1 life cycle was blocked, therefore, we do not think that a dose-dependent analysis of inhibitors is required.

Author response image 1.

(A and B) Representative flow cytometry histograms of VSV-G-pseudotyped HIV-1-EGFP-infected HeLa cells at an MOI of 1, harvested at 48 hpi. The cells were treated with DMSO, the indicated doses of nevirapine (NVP) (A) or indicated doses of raltegravir (RAL) (B) for 24 h before infection.



(1.9) Line 183: Please tell us what ECFP is and why it was chosen. Is there a reference for its failure to form R-loops?

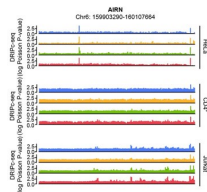
Ibid: The human AIRN gene is a very poor target for HIV integration in PBMC.

A high GC skew value (> 0) is a predisposing factor for R-loop formation at the transcription site. This is because a high GC skew causes a newly synthesized RNA strand to hybridize to the template DNA strand, and the non-template DNA strand remains looped out in a single-stranded conformation (5) (Ref 36 in the revised manuscript). The ECFP sequence possessed a low GC skew value, as previously used for an R-loop-forming negative sequence (6) (Ref 17 of the revised manuscript). We have added this description and the corresponding references to Lines 188–192 of the revised manuscript.

The human AIRN gene (RefSeq DNA sequence: NC_000006.12) sequence possesses a GC skew value of -0.04, in a window centered at base 2186, while the mouse AIRN (mAIRN) sequence is characterized by a GC skew value of 0.213. The ECFP sequence gave a GC skew value of -0.086 in our calculation. We anticipated that the human AIRN gene region does not form a stable R-loop, and in fact, it did not harbor R-loop enrichment upon HIV-1 infection in our DRIPc-seq data analysis of multiple cell types (Author response image 2)

Author response image 2.

Genome browser screenshot over the chromosomal regions in 20-kb windows centered on human AIRN showing results from DRIPc-seq in the indicated HIV-1-infected cells (blue, 0 hpi; yellow, 3 hpi; green, 6 hpi; red, 12 hpi)



(1.10) Line 190: You haven't shown dependence. Associated is a better word.

Thank you for the suggestion. We have changed “R-loop-dependent site-specific HIV-1 integration events...” to “R-loop-associated site-specific HIV-1 integration events...” (Line 198 of the revised manuscript) according to the reviewer’s suggestion in the revised manuscript.

(1.11) Line 239: What happened to P1? What is the relationship of the P and N regions to genes?

We have added superimpositions of the P1 chromatin region on DRIPc-seq and the HIV-1 integration frequency to Figure 4C of the revised manuscript. We observed a relevant integration event within the P1 R-loop region, but to a lesser extent than in the P2 and P3 R-loop regions, perhaps because the P1 region has relatively less R-loop enrichment than the P2 and P3 regions, as examined by DRIP-qPCR in S3A Fig. of the revised manuscript.

Genome browser screenshots with annotations of accommodating genes in the P and N regions are shown in S2A–E Fig. of the revised manuscript, and RNA-seq analysis of the relative gene expression levels of the P1-3 and N1,2 R-loop regions are shown in S4 Table of the revised manuscript. Thank you.

(1.12) Line 261: But the binding affinity of integrase to the R-loop is somewhat weaker than to double-stranded DNA according to Figure 5A.

Nucleic acid substrates were loaded at the same molarity, and the percentage of the unbound fraction was calculated by dividing the intensity of the unbound fraction in each lane by the intensity of the unbound fraction in the lane with 0 nM integrase in the binding reaction. The calculated percentages of the unbound fraction from three independent replicate experiments are shown in Fig. 5A, right of the revised manuscript. In our analysis and measurements, the integrase proteins showed higher binding affinities to the R-loop and R-loop comprising nucleic acid structures than to dsDNA in vitro. We hope that this explanation clarifies this point.

(1.13) Line 337: "accumulate". This is a not uncommon misinterpretation of the results of studies on the distribution of intact proviruses in elite controllers. The only possible correct interpretation of the finding is that proviruses form everywhere else but cells containing them are eliminated, most likely by the immune system.

Thank you for the suggestion. We have changed the Line 337 of the original manuscript to “... HIV-1 proviruses in heterochromatic regions are not eliminated but selected by immune system,” in Lines 361-363 of the revised manuscript.

(1.14) Line 371 How many virus particles per cell does this inoculum amount to?

We determined the amount of GFP reporter viruses required to transduce ~50% of WT Jurkat T cells, corresponding to an approximate MOI of 0.6. We repeatedly obtained 30–50% of VSV-

G-pseudotyped HIV-1-EGFP positively infected cells for HIV1 integration site sequencing library construction for Jurkat T cells.

(1.15) Line 503 and Figures 3 and 4: There must be a clear description of how integration events are quantitated.

Detailed methods for integration site sequencing data processing are described in the Materials and Methods section of the revised manuscript (Line 621–631 of the revised manuscript). We primarily followed HIV-1 integration site sequencing data processing methods previously described in Li et al., mBio, 2020; 11(5) (4).

Reviewer #2 (Public Review):

Retroviral integration in general, and HIV integration in particular, takes place in dsDNA, not in R-loops. Although HIV integration can occur in vitro on naked dsDNA, there is good evidence that, in an infected cell, integration occurs on DNA that is associated with nucleosomes. This review will be presented in two parts. First, a summary will be provided giving some of the reasons to be confident that integration occurs on dsDNA on nucleosomes. The second part will point out some of the obvious problems with the experimental data that are presented in the manuscript.

We appreciate your comments. We have carefully addressed the concerns expressed as follows (your comments are in italics):

(2.1) 2017 Dos Passos Science paper describes the structure of the HIV intasome. The structure makes it clear that the target for integration is dsDNA, not an R-loop, and there are very good reasons to think that structure is physiologically relevant. For example, there is data from the Cherepanov, Engelman, and Lyumkis labs to show that the HIV intasome is quite similar in its overall structure and organization to the structures of the intasomes of other retroviruses. Importantly, these structures explain the way integration creates a small duplication of the host sequences at the integration site. How do the authors propose that an R-loop can replace the dsDNA that was seen in these intasome structures?

We do appreciate the current understanding of the HIV-1 integration site selection mechanism and the known structure of the dsDNA-bound intasome. Our study proposes an R-loop as another contributor to HIV-1 integration site selection. Recent studies providing new perspectives on HIV-1 integration site targeting motivated our current work. For instance, Ajoge et al., 2022 (7) indicated that a guanine-quadruplex (G4) structure formed in the non-template DNA strand of the R-loop influences HIV-1 integration site targeting. Additionally, I. K. Jozwik et al., 2022 (8) showed retroviral integrase protein structure bound to B-to-A transition in target DNA. R-loop structures are a prevalent class of alternative non-B DNA structures (9). We acknowledge the current understanding of HIV-1 integration site selection and explore how R-loop interactions may contribute to this knowledge in the Discussion section of our manuscript.

Primarily based on our current data, we believe that R-loop structures are bound by HIV-1 integrase proteins and lead to HIV-1 viral genome integration into the vicinity regions of the host genomic R-loops, but we do not claim that R-loops completely replace dsDNA as the target for HIV-1 integration. An R-loop can be multi-kilobase in size and the R-loop peak length widely varies depending on the immunoprecipitation and library construction methods (1, 2), therefore, we could not validate whether the center of triple-stranded R-loops is the extraction site of HIV-1 integration where the strand transfer reaction by integrase occurs. Therefore, we replaced phrases such as, “HIV-1 targets R-loops for integration,” which may overstate our finding on the role of R-loop in HIV-1 integration site selection, with

phrases, such as, “HIV-1 favors vicinity regions of R-loop for the viral genome integration,” in the revised manuscript. We apologize for the inconvenience caused by the unclear and non-specific details of our findings. Nevertheless, we believe that we validated R-loop-mediated HIV-1 integration in R-loop-forming regions using our pgR-poor and pgR-rich cell line models. We quantified site-specific integration events in the R-loop regions, and found that a greater number of integration events occurred in designated regions of the pgR-rich cellular genome upon R-loop induction by DOX treatment, but not in pgR-poor cells (Fig. 3E–G of the revised manuscript).

dsDNA may have been the sole target of the intasome demonstrated in vitro possibly because dsDNA has only been considered as a substrate for in vitro intasome assembly. We hope that our work will initiate and advance future investigations on target-bound intasome structures by considering R-loops as potential new targets for integrated proteins and intasomes.

(2.2) As noted above, concerted (two-ended) integration can occur in vitro on a naked dsDNA substrate. However, there is compelling evidence that, in cells, integration preferentially occurs on nucleosomes. Nucleosomes are not found in R loops. In an infected cell, the viral RNA genome of HIV is converted into DNA within the capsid/core which transits the nuclear pore before reverse transcription has been completed. Integration requires the uncoating of the capsid/core, which is linked to the completion of viral DNA synthesis in the nucleus. Two host factors are known to strongly influence integration site selection, CPSF6 and LEDGF. CPSF6 is involved in helping the capsid/core transit the nuclear pore and associate with nuclear speckles. LEDGF is involved in helping the preintegration complex (PIC) find an integration site after it has been released from the capsid/core, most commonly in the bodies of highly expressed genes. In the absence of an interaction of CPSF6 with the core, integration occurs primarily in the lamin-associated domains (LADs). Genes in LADs are usually not expressed or are expressed at low levels. Depending on the cell type, integration in the absence of CPSF6 can be less efficient than normal integration, but that could well be due to a lack of LEDGF (which is associated with expressed genes) in the LADs. In the absence of an interaction of IN with LEDGF (and in cells with low levels of HRP2) integration is less efficient and the obvious preference for integration in highly expressed genes is reduced. Importantly, LEDGF is known to bind histone marks, and will therefore be preferentially associated with nucleosomes, not R-loops. LEDGF fusions, in which the chromatin binding portion of the protein is replaced, can be used to redirect where HIV integrates, and that technique has been used to map the locations of proteins on chromatin. Importantly, LEDGF fusions in which the chromatin binding component of LEDGF is replaced with a module that recognizes specific histone marks direct integration to those marks, confirming integration occurs efficiently on nucleosomes in cells. It is worth noting that it is possible to redirect integration to portions of the host genome that are poorly expressed, which, when taken with the data on integration into LADs (integration in the absence of a CPSF6 interaction) shows that there are circumstances in which there is reasonably efficient integration of HIV DNA in portions of the genome in which there are few if any R-loops.

Although R-loops may not wrap around nucleosomes, long and stable R-loops likely cover stretches of DNA corresponding to multiple nucleosomes (10). For example, R-loops are associated with high levels of histone marks, such as H3K36me3, which LEDGF recognizes (2, 11). R-loops dynamically regulate the chromatin architecture. Possibly by altering nucleosome occupancy, positioning, or turnover, R-loop structures relieve superhelical stress and are often associated with open chromatin marks and active enhancers (2, 10). These features are also distributed over HIV-1 integration sites (12). In the Discussion section of the revised manuscript, we explored the R-loop molding mechanisms in the host genomic environment for HIV-1 integration site selection and its potential collaborative role with LEDGF/p75 and CPSF6 governing HIV-1 integration site selection.

By carefully revising our original manuscript, with respect to the reviewer's comment, we recognized the need to tone down our statements. We found that the title, abstract, and discussion of our original manuscript includes phrases, such as, "HIV-1 targets R-loops for integration," which may overstate our finding on the role of R-loop in HIV-1 integration site selection. We replaced these phrases. For example, we used phrases, such as "HIV-1 favors vicinity regions of R-loop for the viral genome integration," in the revised manuscript. We apologize for the inconvenience caused by the unclear and non-specific details of our findings.

(2.3) Given that HIV DNA is known to preferentially integrate into expressed genes and that R-loops must necessarily involve expressed RNA, it is not surprising that there is a correlation between HIV integration and regions of the genome to which R loops have been mapped. However, it is important to remember that correlation does not necessarily imply causation.

We understand the reviewer's concern regarding the possibility of a coincidental correlation between the R-loop regions and HIV-1 integration sites, particularly when the interpretation of this correlation is primarily based on a global analysis.

Therefore, we designed pgR-poor and pgR-rich cell lines, which we believe are suitable models for distinguishing between integration events driven by transcription and the presence of R-loops. Although the two cell lines showed comparable levels of transcription at the designated region upon DOX treatment via TRE promoter activation (Fig. 3B of the revised manuscript), only pgR-rich cells formed R-loops at the designated regions (Fig. 3C of the revised manuscript). When infected with HIV1, pgR-rich cells, but not pgR-poor cells, showed higher infectivity after DOX treatment (Fig. 3D of the revised manuscript). Moreover, we quantified site-specific integration events in the R-loop regions, and found that a greater number of integration events occurred in designated regions of the pgR-rich cellular genome upon R-loop induction by DOX treatment, but not in pgR-poor cells (Fig. 3E of the revised manuscript). Therefore, we concluded that transcriptional activation without an R-loop (in pgR-poor cells) may not be sufficient to drive HIV-1 integration. We believe that our work provides strong evidence for a causative relationship between R-loop formation/R-loop sites and HIV-1 integration. We hope that our explanation addresses your concerns. Thank you.

If we consider some of the problems in the experiments that are described in the manuscript:

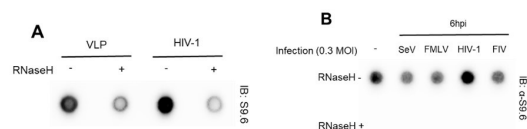
(2.4) In an infected individual, cells are almost always infected by a single virion and the infecting virion is not accompanied by large numbers of damaged or defective virions. This is a key consideration: the claim that infection by HIV affects R-loop formation in cells was done with a VSVg vector in experiments in which there appears to have been about 6000 virions per cell. Although most of the virions prepared in vitro are defective in some way, that does not mean that a large fraction of the defective virions cannot fuse with cells. In normal in vivo infections, HIV has evolved in ways that avoid signaling infected the cell of its presence. To cite an example, carrying out reverse transcription in the capsid/core prevents the host cell from detecting (free) viral DNA in the cytoplasm. The fact that the large effect on R-loop formation which the authors report still occurs in infections done in the absence of reverse transcription strengthens the probability that the effects are due to the massive amounts of virions present, and perhaps to the presence of VSVg, which is quite toxic. To have physiological relevance, the infections would need to be carried out with virions that contain HIV even under circumstances in which there is at most one virion per cell.

Our virus production and in vitro and ex vivo HIV-1 infection experimental conditions, designed for infecting cell types, such as HeLa cells and primary CD4⁺ T cells with VSV-G

pseudotyped HIV, were based on a comprehensive review of numerous references. At the very beginning of this study, we tested HIV-1-specific host genomic R-loop induction using empty virion particles (virus-like particles, VLP) or other types of viruses (non-retrovirus, SeV; retroviruses, FMLV and FIV), all produced with a VSV G protein donor. We could not include a control omitting the VSV G protein or using natural HIV-1 envelope protein to prevent viral spread in culture. We observed that despite all types of virus stocks being prepared using VSV-G, only cells infected with HIV-1 viruses showed R-loop signal enrichment (Author response image 3). Therefore, we omitted the control for the VSV G protein in subsequent analyses, such as DRIPcseq. We have also revised our manuscript to provide a clearer description of the experimental conditions. In particular, we now clearly stated that we used VSV-G pseudotyped HIV-1 in this study, throughout the abstract, results, and discussion sections of the revised manuscript. Thank you.

Author response image 3.

(A) Dot blot analysis of the R-loop in gDNA extracts from HIV-1 infected U2OS cells with MOI of 0.6 harvested at 6 hpi. The gDNA extracts were incubated with or without RNase H in vitro before membrane loading (anti-S9.6 signal). (B) Dot blot analysis of the R-loop in gDNA extracts from HeLa cells infected with 0.3 MOI of indicated viruses. The infected cells were harvested at 6 hpi. The gDNA extracts were incubated with or without RNase H in vitro before membrane loading (anti-S9.6 signal).



HIV-1 co-infection may also be expected in cell-free HIV-1 infections. However, it was previously suggested that the average number of infection events varies within 1.02 to 1.65 based on a mathematical model that estimates the frequency of multiple infections with the same virus (Figure 4c of Ito et al., Sci. Rep, 2017; 6559) (13).

(2.5) Using the Sso7d version of HIV IN in the in vitro binding assays raises some questions, but that is not the real question/problem. The real problem is that the important question is not what/how HIV IN protein binds to, but where/how an intasome binds. An intasome is formed from a combination of IN bound to the ends of viral DNA. In the absence of viral DNA ends, IN does not have the same structure/organization as it has in an intasome. Moreover, HIV IN (even Sso7d, which was modified to improve its behavior) is notoriously sticky and hard to work with. If viral DNA had been included in the experiment, intasomes would need to be prepared and purified for a proper binding experiment. To make matters worse, there are multiple forms of multimeric HIV IN and it is not clear how many HIV INs are present in the PICs that actually carry out integration in an infected cell.

As the reviewer has noted, HIV IN, even with Sso7d tagging, is difficult. We attempted the purification of viral DNA (vDNA)-bound PICs using either Sso7d-tagged HIV-1 integrase proteins or non-tagged HIV-1 integrase proteins (F185K/C280S), procured from the NIH HIV reagent program (HRP-20203), following the method described by Passos et al., Science, 2017; 355 (89-92) (3). Despite multiple attempts, we were unable to purify the vDNA-bound IN protein complexes for in vitro assays. However, through multiple biochemical experiments, we believe that we have successfully demonstrated the interaction between cellular R-loops and HIV-1 integrase proteins both in cells and in vitro (Fig. 5A–F of the revised manuscript). We also observed a close association between integrase proteins and host cellular R-loops in

HIV-1-infected cells, using a fluorescent recombinant virus (HIV-IN-EGFP) with intact IN-EGFP PICs (Fig. 5G of the revised manuscript).

(2.6) As an extension of comment 2, the proper association of an HIV intasome/PIC with the host genome requires LEDGF and the appropriate nucleic acid targets need to be chromatinized.

The interaction between cellular R-loops and HIV-1 integrase proteins in HeLa cells endogenously expressing LEDGF/p75 was examined using reciprocal immunoprecipitation assays in Fig. 5C–F, S6B, and S6D Fig. of the revised manuscript. In addition, as discussed in more detail in our response to comment [28], we observed a close association between host cellular R-loops and HIV-1 integrase proteins by PLA assay, in HIV-1-infected HeLa cells.

(2.7) Expressing any form of IN, by itself, in cells to look for what IN associates with is not a valid experiment. A major factor that helps to determine both where integration takes place and the sites chosen for integration is the transport of the viral DNA and IN into the nucleus in the capsid core. However, even if we ignore that important part of the problem, the IN that the authors expressed in HeLa cells won't be bound to the viral DNA ends (see comment 2), even if the fusion protein would be able to form an intasome. As such, the IN that is expressed free in cells will not form a proper intasome/PIC and cannot be expected to bind where/how an intasome/PIC would bind.

As discussed in more detail in our response to comment [2-8], we believe that our PLA experiment using the pVpr-IN-EGFP virus, which has previously been examined for virion integrity, as well as the IN-EGFP PICs (14), demonstrated a close association between host cellular R-loops and HIV-1 integrase proteins in HIV-1-infected cells.

(2.8) As in comment 1, for the PLA experiments presented in Figure 5 to work, the number of virions used per cell (which differs from the MOI measured by the number of cells that express a viral marker) must have a high, which is likely to have affected the cells and the results of the experiment. However, there is the additional question of whether the IN-EGFP fusion is functional. The fact that the functional intasome is a complex multimer suggests that this could be a problem. There is an additional problem, even if IN-GFP is fully functional. During a normal infection, the capsid core will have delivered copies of IN (and, in the experiments reported here, the IN-GFP fusion) into the nucleus that is not part of the intasome. These "free" copies of IN (here IN-GFP) are not likely to go to the same sites as an intasome, making this experiment problematic (comment 4).

The HIV-IN-EGFP virus stock was produced by polyethylenimine-mediated transfection of HEK293T cells with 6 µg of pVpr-IN-EGFP, 6 µg of HIV-1 NL4-3 noninfectious molecular clone (pD64E; NIH AIDS Reagent Program 10180), and 1 µg of pVSV-G as previously described in (14), and described in the Materials and Methods section of our manuscript. The pVpr-IN-EGFP vector used to produce HIV-1-IN-EGFP virus stock was provided by Anna Cereseto group (Albanese et al., PLOS ONE, 2008; 6(6); Ref 34 of the revised manuscript). It was previously reported that the HIV-1IN-EGFP virions produced by IN-EGFP trans-incorporation through Vpr are intact and infective viral particles (Figure 1 of Albanese et al., PLOS ONE, 2008; 6(6)). Therefore, we believe that the HIV-IN-EGFP used in our PLA experiments was functional.

Additionally, Albanese et al. showed that the EGFP signal of HIV-IN-EGFP virions colocalizes with the viral protein matrix (p17MA) and capsid (P24CA) as well as with the newly synthesized cDNA produced by reverse transcriptase by labeling and visualizing the synthesized cDNA (14). In addition, the fluorescent recombinant virus (HIV-IN-EGFP) was structurally intact at the nuclear level (Figure 6 of Albanese et al., PLOS ONE, 2008; 6(6)).

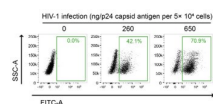
Therefore, we believe that our PLA experimental result is not likely misled as the reviewer concerns due to the integrity of the HIV-IN-EGFP virion as well as IN-EGFP PICs.

Furthermore, the in vitro HIV-1 infection setting of our PLA experiments was carefully determined based on multiple studies that performed image-based assays on HIV-1-infected cells. For instance, Albanese et al. infected 4×10^4 cells with viral loads equivalent to 1.5 or 3 μ g of HIV-1 p24 for their immunofluorescence analysis, in their previous report (14). We titrated the fluorescent HIV-1 virus stocks by examining both the multiplicity of infection (MOI) and quantifying the HIV-1 p24 antigen content (Author response image 4). In our calculation, we infected 5×10^4 HeLa cells with viral loads equivalent to 1.3 μ g of HIV-1 p24, which is indicated as 2 MOI in Fig. 5G of our manuscript, for our PLA experiments.

Image-Based Assays often require increased and enhanced signal for statistical robustness. For example, Achuthan et al. infected cells with VSV-G-pseudotyped HIV1 at the approximate MOI of 350 for vDNA and PIC visualization (15). Therefore, we believe our experimental condition for PLA experiments, which we carefully designed based on previous study that are frequently referred, are reasonable. We really hope that our discussion sufficiently addressed the reviewer's concern.

Author response image 4.

Gating strategy used to determine HIV-1-infectivity in HeLa cells at 48 hpi. Cells were infected with a known p24 antigen content in the stock of the VSV-G-pseudotyped HIV-1-EGFP-virus. The percentages of GFP-positive cell population are indicated.



(2.9) In the Introduction, the authors state that the site of integration affects the probability that the resulting provirus will be expressed. Although this idea is widely believed in the field, the actual data supporting it are, at best, weak. See, for example, the data from the Bushman lab showing that the distribution of integration sites is the same in cells in which the integrated proviruses are, and are not, expressed. However, given what the authors claim in the introduction, they should be more careful in interpreting enzyme expression levels (luciferase) as a measure of integration efficiency in experiments in which they claim proviruses are integrated in different places.

We thank the reviewer for the constructive comment. We have changed the statement in Lines 41–42 in the Introduction section of our original manuscript to “The chromosomal landscape of HIV-1 integration influences proviral gene expression, persistence of integrated proviruses, and prognosis of antiretroviral therapy.” (Lines 39–41 of the revised manuscript). We believe that this change can tone-down the relevance between the site of integration and the provirus expression level.

The piggyBac transposase randomly insert the “cargo (transposon)” into TTAA chromosomal sites of the target genome, generating efficient insertions at different genomic loci (16, 17). We believe that this random insertion of the pgR-poor/rich vector mediated by the piggyBac system allows us not to mislead the R-loop-mediated HIV1 integration site because of the genome locus bias of the vector insertion. Therefore, Figure 3 in our manuscript does not claim any relevance between the site of integration and the resulting provirus expression levels. Instead, as noted in Line 214 of the revised manuscript, using the luciferase reporter HIV-1 virus, we attempted to examine HIV-1 infection in cells with an “extra number of R-loops” in the host cellular genome. We observed that pgR-rich cells showed higher luciferase

activity upon DOX treatment than pgR-poor cells (Fig. 3D of the revised manuscript). We believe that this is because a greater number of HIV-1 integration events may occur in pgR-rich cells, where DOX-inducible de novo R-loop regions are introduced. This has been further examined in Fig. 3E–G of the revised manuscript. We hope this explanation clarifies the Figure 3. Thank you.

(2.10) Using restriction enzymes to create an integration site library introduces biases that derive from the uneven distribution of the recognition sites for the restriction enzymes.

As described in the Materials and Methods section, we adopted a sequencing library construction method using a previously established protocol (18, 19). Although we recognize the advantages of DNA fragmentation by sonication, in in vitro or ex vivo HIV-1 infection settings, where the multiplicity of infection is carefully determined based on multiple references, more copies of integrated viral sequences are expected compared to that in samples from infected patients (18). Therefore, in these settings, restriction enzyme-based DNA fragmentation and ligation-mediated PCR sequencing are well-established methods that provide significant data sources for HIV-1 integration site sequencing (15, 20–22). Furthermore, our data showing the proportion of integration sites over R-loop regions (Fig. 4B of the revised manuscript) are presented alongside the respective random controls (i.e., proportion of integration sites within the 30-kb windows centered on randomized DRIPc-seq peaks, gray dotted lines; control comparisons between randomized integration sites with DRIPc-seq peaks, black dotted lines; and randomized integration sites with randomized DRIPc-seq peaks, gray solid lines), which do not show such a correlation between the HIV-1 integration sites and nearby areas of the R-loop regions. Therefore, we believe that our results from the integration site sequencing data analysis are unlikely to be biased.

Reviewer #3 (Public Review):

In this manuscript, Park and colleagues describe a series of experiments that investigate the role of R-loops in HIV-1 genome integration. The authors show that during HIV-1 infection, R-loops levels on the host genome accumulate. Using a synthetic R-loop prone gene construct, they show that HIV-1 integration sites target sites with high R-loop levels. They further show that integration sites on the endogenous host genome are correlated with sites prone to R-loops. Using biochemical approaches, as well as in vivo co-IP and proximity ligation experiments, the authors show that HIV-1 integrase physically interacts with R-loop structures.

My primary concern with the paper is with the interpretations the authors make about their genome-wide analyses. I think that including some additional analyses of the genome-wide data, as well as some textual changes can help make these interpretations more congruent with what the data demonstrate. Here are a few specific comments and questions:

We are grateful for the time and effort we spent on our behalf and the reviewer's appreciation for the novelty of our work, in particular, R-loop induction by HIV-1 infection and the correlation between host R-loops and the genomic site of HIV-1 integration. In the following sections, we provide our responses to your comments and suggestions. Your comments are in italics. We have carefully addressed the following issues.

(3.1) I think Figure 1 makes a good case for the conclusion that R-loops are more easily detected HIV-1 infected cells by multiple approaches (all using the S9.6 antibody). The authors show that their signals are RNase H sensitive, which is a critical control. For the DRIPc-Seq, I think including an analysis of biological replicates would greatly strengthen the manuscript. The authors state in the methods that the DRIPc pulldown experiments

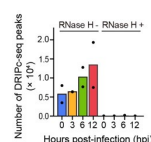
were done in biological replicates for each condition. Are the increases in DRIPc peaks similar across biological replicates? Are genomic locations of HIV-1-dependent peaks similar across biological replicates? Measuring and reporting the biological variation between replicate experiments is crucial for making conclusions about increases in R-loop peak frequency. This is partially alleviated by the locus-specific data in Figure S3A. However, a better understanding of how the genome-wide data varies across biological replicates will greatly enhance the quality of Figure 1.

DRIPc-seq experiments were conducted with two biological replicates. To define consensus DRIPc-seq peaks using these two replicates, we used two methods applicable to ChIP-seq analysis: the irreproducible discovery rate (IDR) method and sequencing data pooling. We found that the sequencing data pooling method yielded significantly more DRIPc-seq peaks than consensus peak identification through IDR, and we decided to utilize R-loop peaks from pooled sequencing data for our downstream analyses, as described in the figure legends and Materials and Methods of the revised manuscript.

As noted by the reviewer, it is important to verify whether the increasing trend in the number of R-loop peaks and genomic locations of HIV-1 dependent R-loops were consistently observed across the two biological replicates. Therefore, we independently performed R-loop calling on each replicate of the sequencing data of primary CD4+ T cells from two individual donors to verify that the increase in R-loop numbers was consistent (Author response image 5). Additionally, the overlap of the R-loop peaks between the two replicates was statistically significant across the genome (Author response table 1). Thank you.

Author response image 5.

Bar graph indicating DRIPc-seq peak counts for HIV-1-infected primary CD4+ T cells harvested at the indicated hours post infection (hpi). Pre-immunoprecipitated samples were untreated (–) or treated (+) with RNase H, as indicated. Each dot corresponds to an individual data set from two biologically independent experiments.



Author response table 1.

DRIPc-seq peak length and Chi-square p-value in CD4+ T cells from individual donor 1 and 2

	Total DRIPc-seq peak length (replicate 1, bp)	Total DRIPc-seq peak length (replicate 2, bp)	Total DRIPc-seq peak length (overlap, bp)	Chi-square p-value
0hpi	1910609	4722514	540640	p < 1e-308
3hpi	3630859	3308270	835457	p < 1e-308
6hpi	4195887	8496266	1485568	p < 1e-308
12hpi	4323824	13714402	1903473	p < 1e-308

(3.2) I think that the conclusion that R-loops "accumulate" in infected cells is acceptable, given the data presented. However, in line 134 the authors state that "HIV1 infection induced host genomic R-loop formation". I suggest being very specific about the observation. Accumulation can happen by (a) inducing a higher frequency of the occurrence of individual R-loops and/or (b) stabilizing existing R-loops. I'm not convinced the authors present enough evidence to claim one over the other. It is altogether possible that HIV-1 infection stabilizes R-loops such that they are more persistent (perhaps by

interactions with integrase?), and therefore more easily detected. I think rephrasing the conclusions to include this possibility would alleviate my concerns.

We thank the reviewer for the considerable discussion on our manuscript. We have now changed Line 134 to, “HIV-1 infection induces host genomic R-loop enrichment” (Lines 132–133 of the revised manuscript), and added a new conclusion sentence implicating the possible explanation for the R-loop signal enrichment upon HIV-1 infection (Lines 133–135 of the revised manuscript), according to the reviewer's suggestion.

(3.3) A technical problem with using the S9.6 antibody for the detection of R-loops via microscopy is that it cross-reacts with double-stranded RNA. This has been addressed by the work of Chedin and colleagues (as well as others). It is absolutely essential to treat these samples with an RNA:RNA hybrid-specific RNase, which the authors did not include, as far as their methods section states. Therefore, it is difficult to interpret all of the immunofluorescence experiments that depend on S9.6 binding.

We understand the reviewer's concern regarding the cross-reactivity of the S9.6 antibody with more abundant dsRNA, particularly in imaging applications. We carefully designed the experimental and analytical methods for R-loop detection using microscopy. For example, we pre-extracted the cytoplasmic fraction before staining with the S9.6 antibody and quantified the R-loop signal by subtracting the nucleolar signal. Both of these steps were taken to eliminate the possibility of misdetecting R-loops via microscopy because of the prominent cytoplasmic and nucleolar S9.6 signals, which primarily originate from ribosomal RNA. In addition, we included R-loop negative control samples in our microscopy analysis that were subjected to intensive RNase H treatment (60U/mL RNase H for 36 h) and observed a significant reduction in the S9.6 signal (Figure 1E of the revised manuscript). RNase H-treated samples served as essential and widely accepted negative controls for R-loop detection.

We would like to point out that recent studies have reported strong intrinsic specificity of S9.6 antibody for DNA:RNA hybrid duplex over dsDNA and dsRNA, along with the structural elucidations of S9.6 antibody recognition of hybrids (23, 24). Therefore, our interpretation of host cellular R-loop enrichment after HIV-1 infection using S9.6 antibodies in multiple biochemical approaches is well supported. Nevertheless, we agree with the reviewer's opinion that additional negative controls for the detection of R-loops via microscopy, such as RNase T1- and RNase III-treated samples, could improve the robustness and accuracy of R-loop imaging data (25).

(3.4) Given that there is no clear correlation between expression levels and R-loop peak detection, combined with the data that show increased detection of R-loop frequency in non-genic regions, I think it will be important to show that the R-loop forming regions are indeed transcribed above background levels. This will help alleviate possible concerns that there are technical errors in R-loop peak detection.

Figures S5D and S5E in the revised manuscript show the relative gene expression levels of the R-loop-forming positive regions (P1-3) and the referenced R-loop-positive loci (RPL13A and CALM3). The gene expression levels of these R-loop-forming regions were significantly higher than those of the ECFP or mAIRN genes without DOX treatment, which can be considered background levels of transcription in cells. Thank you.

(3.5) In Figures 4C and D the hashed lines are not defined. It is also interesting that the integration sites do not line up with R-loop peaks. This does not necessarily directly refute the conclusions (especially given the scale of the genomic region displayed), but should be addressed in the manuscript. Additionally, it would greatly improve Figure 4 to have some idea about the biological variation across replicates of the data presented 4A.

We thank the reviewer for the considerable comment on our study. First of all, we added an annotation for the dashed lines in the figure legends of Figures 4C and 4D in the revised manuscript.

We agree with the reviewer's interpretation of the relationship between the integration sites and R-loop peaks. Primarily based on our current data, we believe R-loop structures are bound by HIV-1 integrase proteins and lead HIV-1 viral genome integration into the “vicinity” regions of the host genomic R-loops. We displayed a large-scale genomic region (30-kb windows) to present integration sites surrounding R-loop centers because an R-loop can be multi-kilobase in size (1, 2). Depending on the immunoprecipitation and library construction methods, the R-loop peaks varied in size, and the peak length showed a wide distribution (Figure 3B of Malig et al., 2020, Figure 1B of Sanz et al., 2016, and Figure 2A of the revised manuscript). Therefore, presenting integration site events within a wide window of R-loop peaks could be more informative and better reflect the current understanding of R-loop biology.

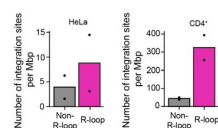
R-loop formation recruits diverse chromatin-binding protein factors, such as H3K4me1, p300, CTCF, RAD21, and ZNF143 (Figure 6A and 6B of Sanz et al., 2016) (26), which allow R-loops to exhibit enhancer and insulator chromatin states, which can act as distal regulatory elements (26, 27). We have demonstrated physical interactions between host cellular R-loops and HIV-1 integrase proteins (Figure 5 of the revised manuscript), therefore, we believe that this ‘distal regulatory element-like feature’ of the R-loop can be a potential explanation for how R-loops drive integration over longrange genomic regions.

According to your suggestion, we added this explanation to the relevant literature in the Discussion section of the revised manuscript.

Author response image 6 which represents the biological variation across replicates of the data shown in Figure 4A. The integration site sequencing data for Jurkat cells were adopted from SRR12322252 (4), which consists of the integration site sequencing data of HIV-1-infected wild type Jurkat cells with one biological replicate. We hope that our explanations and discussion have successfully addressed your concerns. Thank you.

Author response image 6.

Bar graphs showing the quantified number of HIV-1 integration sites per Mb pair in total regions of 30-kb windows centered on DRIP-seq peaks from HIV-1 infected HeLa cells and primary CD4⁺ T cells (magenta) or non-R-loop region in the cellular genome (gray). Each dot corresponds to an individual data set from two biologically independent experiments.



(3.6) The authors do not adequately describe the Integrase mutant that they use in their biochemical experiments in Figure 5A. Could this impact the activity of the protein in such a way that interferes with the interpretation of the experiment? The mutant is not used in subsequent experiments for Figure 5 and so even though the data are consistent with each other (and the conclusion that Integrase interacts with R-loops) a more thorough explanation of why that mutant was used and how it impacts the biochemical activity of the protein will help the interpretation of the data presented in Figure 5.

We appreciate the reviewer's suggestions. In our EMSA analysis, we purified and used Sso7d-tagged HIV-1 integrase proteins with an active-site amino acid substitution, E152Q. First, we used the Sso7d-tagged HIV-1 integrase protein, as it has been suggested in previous studies that the fusion of small domains, such as Sso7d (DNA binding domain) can significantly improve the solubility of HIV integrase proteins without affecting their ability to assemble with substrate nucleic acids and their enzymatic activity (Figure 1B of Li et al., PLOS ONE, 2014;9 (8) (28, 29). We used an integrase protein with an active site amino acid substitution, E152Q, in our mobility shift assay, because the primary goal of this experiment was to examine the ability of the protein to bind or form a complex with different nucleic acid substrates. We thought that abolishing the enzymatic activity of the integrase protein, such as 3'-processing that cleaves DNA substrates, would be more appropriate for our experimental objective. This Sso7d tagged- HIV-1 integrase with the E152Q mutation has also been used to elucidate the structural model of the integrase complex with a nucleic acid substrate by cryo-EM (3) and has been shown to not disturb substrate binding. Based on the reviewer's comments, we have added a description of the E152Q mutant integrase protein in Lines 268–270 of the revised manuscript. Thank you.

Reviewer #3 (Recommendations For The Authors):

The paper suffers from many grammatical errors, which sometimes interfere with the interpretations of the experiments. In the view of this reviewer, the manuscript must be carefully revised prior to publication. For example, lines 247-248 "Intasomes consist of HIV-1 viral cDNA and HIV-1 coding protein, integrases." It is unclear from this sentence whether there are multiple integrases or multiple proteins that interact with the viral genome to facilitate integration. This makes the subsequent experiments in Figure 5 difficult to interpret. There are many other examples, too numerous to point out individually.

We thoughtfully revised the original manuscript, making the best efforts to provide clearer details of our findings. We believe that we have made substantial changes to the manuscript, including Lines 247–248 of the original manuscript that the reviewer noted. Furthermore, the revised manuscript was edited by a professional editing service. Thank you.

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<https://doi.org/10.7554/eLife.97348.2.sa0>